

Research on the Estimation of Elderly Care
Needs and Strategy for Caregiver Team
Construction in Xuhui District

School of International and Public Affairs, Shanghai Jiao Tong University

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Abstract

The "15th Five-Year Plan" period is a critical window of opportunity for Xuhui District to address deep population aging and refine its elderly care service supply system. In order to scientifically formulate the development plan for the elderly caregiver workforce, this study comprehensively utilized the Xuhui District Statistical Yearbook, the Shanghai Elderly Care Service Statistical Information, and survey data provided by 13 sub-districts and towns across the district. A five-level cascaded calculation model of "Elderly Population – Disability Status – Care Hours – Caregiver Demand – Supply-Demand Gap" was constructed. By adopting three scenario envelopes (optimistic, baseline, and prudent), this study clarified the scale of net demand for full-scope formal care across the district, systematically calculated the demand scale and supply-demand gap of caregivers corresponding to the registered elderly population in Xuhui District, and provided quantitative decision-making basis for the caregiver workforce construction in the whole district.

The core findings of this study include:

First, the elderly population is rising steadily, the trend of advanced aging and disability is prominent, and elderly care service demands are polarizing. In the baseline scenario, the registered population aged 60 and above in the district will achieve a net increase of approximately 12,500 persons (with an average annual growth rate of 0.87%) over the next five years. Among them, the proportion of the elderly population aged 80 and above will rise to 18.67% in 2030, ranking first in the city in terms of the degree of advanced aging. Accompanied by advanced aging is the rigid expansion of the disabled population. By 2030, the total number of disabled elderly individuals in the district will reach 45,400, and the severely disabled group, which accounts for only 13.97%, consumes nearly 40% of the district's total annual care hours. Surveys indicate that ordinary elderly individuals have a high willingness to pay for basic and inclusive services such as medical care, nutrition, and meal assistance, while their actual consumption is concentrated on domestic cleaning and daily living care; well-educated and high-net-worth elderly individuals pursue high-quality, personalized care that demonstrates spiritual dignity and values autonomy over life, yet they tend to address needs through private channels due to privacy barriers. In addition, community and home-based care needs exhibit characteristics of high frequency, fragmentation, and heavy labor dependence, with minimal demand for specialized medical nursing.

Second, the overall supply-demand gap of caregivers is controllable, but the labor replenishment and personnel demand in community and home-based care scenarios are extremely high. Taking the data at the end of 2025 as the calculation baseline period and supply

base, and assuming the current service proportions remain unchanged, the demand for elderly care institution caregivers in 2030 will be 1,251 persons, with a cumulative supply-demand gap of only 114 persons; the demand in community and home-based scenarios will reach 834 persons, with a cumulative supply-demand gap as high as 414 persons, nearly doubling in number. Front-line service records reveal that authentic demands in community and home-based settings are heavily concentrated on domestic cleaning, daily living care and bathing assistance, rehabilitation and functional capability training, and meal assistance services, demonstrating characteristics of high fragmentation, high frequency, and heavy labor dependence. Simultaneously, certain high-quality, personalized specialized services are also being gradually promoted, such as cognitive ability maintenance, health management, nutritional guidance, and cultural companionship. A stable client base that recognizes professional value and is willing to pay for service quality has been accumulated, providing practical support for upgrading the service system from a basic security type to a quality enhancement type.

Third, the caregiver team faces structural imbalances, and professionalism and stability urgently require institutional breakthroughs. The average age of in-service caregivers in the district has exceeded 52 years, among whom those aged 50 and above account for nearly 80%, and females account for 94.48%. The pattern of "younger elderly caring for older elderly" triggers risks of workforce disruption. The educational attainment of the workforce is generally low, and the skill structure exhibits typical bottom-heavy sedimentation characteristics, making it difficult to cope with complex and fine-grained care for disabled and cognitively impaired seniors. Furthermore, the average monthly salary in the industry is approximately 5,300 yuan and piece-rate wage systems are prevalent; over 60% of practitioners work an average of 9 to 12 hours or more per day. Squeezed doubly by overtime, high-intensity labor, and high housing and living costs in the central urban area, the annual caregiver turnover rate hovers at a high level of 50% to 60%.

In response to the aforementioned supply-demand contradictions and realistic bottlenecks, the construction of the elderly caregiver workforce in the district must adhere closely to the four collaborative pillars of "Silver Economy, Digital and Intelligent Empowerment, Chain Integration, and Basic Safety Net", specifically including: First, activate new momentum for the silver economy and categorically guide multi-tiered elderly care consumption demands. Second, adhere to both quantitative expansion and quality improvement, comprehensively implementing technology empowerment and the "Digital Intelligence – Embodied Coupling" strategy. Third, connect the "Institution – Community – Home" service chain and promote the precise allocation of professional care resources toward community and home settings. Fourth, improve the comprehensive security system for front-line caregivers and adopt multiple measures to alleviate labor contradictions and living pressures.

Table of Contents

Abstract	2
I. Research Background.....	5
II. Analysis of Care Needs and Caregiver Supply Structure in Xuhui District During the "15th Five-Year Plan" Period.....	5
2.1 Aging Steadily Rises, and Elderly Care Service Needs Differentiate	5
2.2The Trend of Advanced Aging Is Obvious, and the Demand for Severe Disability Care Is High	8
2.3 The Supply-Demand Gap of Caregivers Is Generally Controllable, but the Demand for Community and Home-Based Services Is Extremely High	9
2.4 The Professionalism and Stability of the Caregiver Workforce Urgent for Institutional Breakthroughs.....	12
III. Countermeasure Suggestions on the Construction of the Elderly Caregiver Team in Xuhui District	14
3.1 Activate New Momentum of Silver Economy and Categorically Guide Multi-Level Elderly Care Consumption Needs	14
3.2 Adhere to Both Scale Increase and Quality Improvement, and Comprehensively Enhance Professional Quality of Fine Care for Disability and Quality Services	14
3.3 Connect the "Institution-Community-Home" Service Chain and Promote the Precise Leaning of Professional Care Resources toward Community and Home Care	16
3.4 Improve the Comprehensive Security System for Front-Line Caregivers and Take Multiple Measures to Alleviate Employment Contradictions and Living Cost Pressures	17

I. Core Issues of the Study

Addressing the practical needs of elderly caregiver workforce construction in Xuhui District during the "15th Five-Year Plan" period, this study systematically answers four progressively interlocking core questions: the evolution of the population base, the scale and composition of disability, the supply-demand matching gap of caregivers, and the development path of the caregiver workforce. Answering these four questions is not only a practical requirement for the Xuhui District People's Government in formulating the "15th Five-Year Plan" special program for elderly care services and the caregiver team construction plan, but also serves as a methodological sample and policy practice reference for other deeply aging central urban areas in Shanghai City and nationwide.

Prior to carrying out the calculation work, the research team organized field investigations across various sub-districts in Xuhui District, providing support for subsequent calculation parameter settings and policy recommendations. This study constructed a five-level cascaded calculation model comprising population, disability, care, demand, and gap, taking the registered elderly population of Xuhui District as the starting point and deriving the supply-demand gap through five progressive links. Each link is closely connected, achieving complete traceability from input parameters to final outcomes; for details, please refer to the *Technical Specification Document for the Calculation of the Number of Disabled Elderly and the Shortage of Elderly Caregivers in Xuhui District* and the *Technical Specification for the Estimation and Prediction of the Elderly Population in Shanghai and Xuhui District*. This study distinguishes the entities supplying care services and converts the overall caregiver demand into actual team construction demand scales for the two major service supply entities: elderly care institutions and communities.

II. Analysis of Care Needs and Caregiver Supply Structure in Xuhui District During the "15th Five-Year Plan" Period

2.1 Aging Steadily Rises, and Elderly Care Service Needs Differentiate

First, during the "15th Five-Year Plan" period, the registered elderly population aged 60 and above in Xuhui District will demonstrate a steady growth trend in total volume. According to the multi-scenario cascaded calculation model forecasts, under the baseline scenario, it is estimated that the registered elderly population will grow steadily from 354,400 persons in 2026 to 366,900 persons in 2030, with a net increase of approximately 12,500 persons over five years, representing an average annual growth rate of 0.87%. The degree of population aging in the district remains at a medium-low level among all districts across the municipality, but the absolute volume of demand for elderly care services remains very substantial and will continue to expand; see the

appendix *Results of Population Estimation and Prediction of the Elderly in Shanghai and Xuhui* for details.

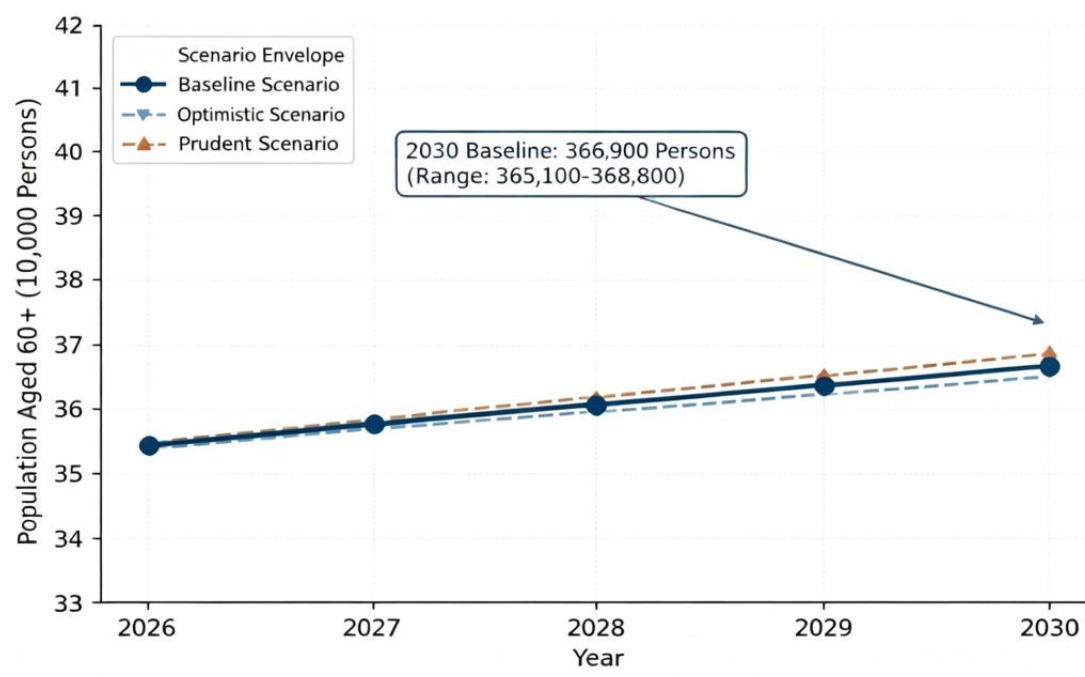


Figure 1 Evolution Trend of Elderly Population Aged 60 and Above in Xuhui District

Second, the elderly care consumption preference structure of the senior population displays a bipolar differentiation. According to special survey data from the team, **the elderly population exhibits a relatively high willingness to pay for basic and inclusive services that satisfy basic living and health needs**, including medical care, nutrition and healthcare, community meal assistance, physical examination, tourism, home care and care products, senior clubs, and senior education. The specific proportions of willingness to pay are shown in Table 1 below:

Table 1 Elderly Care Products and Services with Relatively High Willingness to Pay Among the Elderly

Elderly Care Products and Service Items	Willingness to Pay Proportion
Medical Care	94.97%
Nutrition and Health Care	92.67%
Community Meal Assistance	80.43%
Physical Examination	78.75%
Tourism	72.65%
Home Care and Care Products	65.44%

Elderly Care Products and Service Items	Willingness to Pay Proportion
Senior Clubs and Senior Education	61.58%

In sharp contrast, however, **the elderly's willingness to pay for improvement-oriented services and services that do not directly enhance immediate well-being is noticeably low**, including: smart home products, home aging-friendly modifications, health monitoring and emergency call products, institutional elderly care, and financial and wealth management professional services. Specific willingness to pay is shown in Table 2 below:

Table 2 Elderly Care Products and Services with Relatively Low Willingness to Pay Among the Elderly

Elderly Care Products and Service Items	Willingness to Pay Proportion
Housekeeping	40.83%
Rehabilitation Training	40.72%
Smart Home Products	30.26%
Home Aging-Friendly Modification	22.37%
Health Monitoring and Emergency Call Products	22.26%
Institutional Elderly Care	15.10%
Financial and Wealth Management Professional Services	7.21%

Third, well-educated and high-net-worth elderly groups in Xuhui District demonstrate prominent characteristics of high personalization and autonomy regarding elderly care service demands. As the core area of science, education, and culture in Shanghai City, Xuhui District brings together a substantial number of university scholars, scientific research institute experts, and retired administrative cadres. The demand of this group for elderly care services is leaping toward the level of "high quality, personalization, and manifestation of spiritual dignity". At the same time, strong trust and privacy barriers exist among well-educated and high-net-worth elderly groups, leading to a significantly low conversion rate from nominal demand to effective service demand. Finally, this elderly group attaches greater importance to safeguarding life autonomy, fine conditioning of health capital, and the continuation of cognitive and social

capabilities, and is willing to pay for customized, high-quality elderly care products that are genuinely professional, reciprocal, and culturally resonant. These characteristics indicate that when coordinating and promoting caregiver team construction, Xuhui District must abandon traditional equalized supply thinking. While ensuring a baseline for basic needs, it must enhance precision customized service capabilities and actively open up a differentiated development path catering to the needs of high-quality elderly individuals.

2.2 The Trend of Advanced Aging Is Obvious, and the Demand for Severe Disability Care Is High

First, during the "15th Five-Year Plan" period, the proportion of elderly people aged 80 and above among the elderly population aged 60 and above in this district will continue to rise, maintaining the highest rank in the city. Specifically, this proportion will increase from 18.00% at the end of 2025 to 18.67% in 2030. The concentration of advanced-age seniors in this district is closely linked to the gathering of high-quality medical resources within the district, but it also poses challenges to the development of professional care resources in the district.

Second, the total number of disabled elderly people in the district will continue to climb with advanced aging. Under the baseline scenario, the number of disabled elderly people in Xuhui District will be 42,200 in 2026 and will reach 45,400 in 2030. The number of disabled elderly will experience a net increase of 3,200 persons over the next five years, representing an average annual growth of approximately 1.82%. In terms of disability severity, the relative proportions of the three tiers (mild, moderate, and severe) will remain basically stable over the five years, **among which severe disability will increase from approximately 5,900 persons in 2026 to approximately 6,300 persons in 2030; moderate disability will increase from approximately 13,100 persons to approximately 14,100 persons; and mild disability will increase from approximately 23,200 persons to approximately 24,900 persons.**

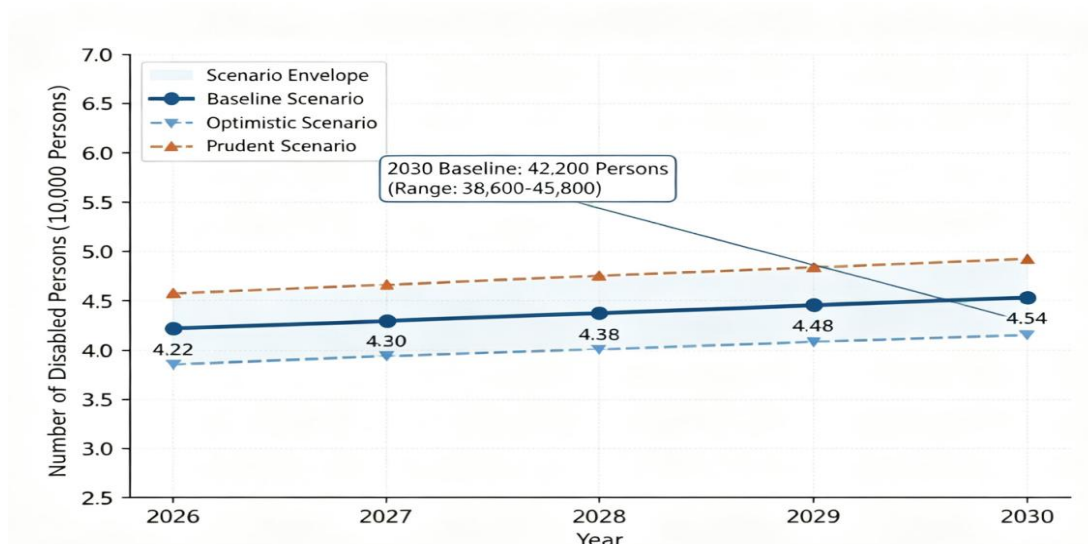


Figure 2 Trend of the Number of Disabled Elderly People Under Different Scenarios in Xuhui District

Third, the care manpower demand for severely disabled elderly accounts for the highest proportion. In the baseline scenario, the standard care hours required for disabled elderly in this district in 2026 is 34,748,000 hours (3,474.80 ten-thousand hours), and will reach 37,348,600 hours (3,734.86 ten-thousand hours) in 2030. Among them, by 2030, the severely disabled group will require approximately 18,503,800 hours (1,850.38 ten-thousand hours), accounting for 50%; the moderately disabled group will require 15,432,500 hours (1,543.25 ten-thousand hours), accounting for 41%; the mildly disabled group requires care hours of approximately 3,412,300 hours (341.23 ten-thousand hours), accounting for 9%, as detailed in Table 3.

Table 3 Number of Elderly People with Different Degrees of Disability and Care Hours in 2030

	Mild	Proportion	Moderate	Proportion	Severe	Proportion
Disabled Elderly (10,000 Persons)	2.49	55%	1.41	31%	0.63	14%
Care Hours (10,000 Hours)	341.23	9%	1543.25	41%	1850.38	50%

2.3 The Supply-Demand Gap of Caregivers Is Generally Controllable, but the Demand for Community and Home-Based Services Is High

First, in the baseline scenario, it is estimated that by 2030, this district will have a caregiver shortage of 114 persons in institutional services, and a caregiver shortage of 414 persons in community and home-based services. The benchmark for this calculation is the registered number of caregivers owned by the whole district as of the end of 2025 published by the government, namely: 1,137 institutional elderly care caregivers, and 420 community and

home-based elderly care caregivers. This result indicates that during the "15th Five-Year Plan" period, the caregiver shortage in this district is generally controllable; for details, refer to the annex *Results of Calculation on the Number of Disabled Elderly and Elderly Caregiver Gap in Xuhui District*.

Among them, under the condition that the supply proportion of various care service types remains unchanged, by the end of 2030, the total demand for institutional elderly care caregivers in the whole district will rise to 1,251 persons, which is 114 persons more than the 1,137 registered institutional caregivers actually existing in this district at the end of 2025. The predicted demand numbers and the differences from the end of 2025 for each year are shown in the table below:

Table 4 Projected Demand for Elderly Care Institution Caregivers in Xuhui District During the "15th Five-Year Plan" Period

Year	Projected Demand for Institutional Caregivers (Persons)	Difference from Actual Number at the End of 2025 (Persons)
2026	1,163	26
2027	1,186	49
2028	1,208	71
2029	1,229	92
2030	1,251	114

Among them, under the condition that the supply proportion of various care service types remains unchanged, by the end of 2030, the total demand for community and home-based caregivers in the whole district will rise to 834 persons, which is 414 persons more than the registered actual community and home-based caregivers at the end of 2025, nearly doubling in number. Therefore, how to precisely fill these position gaps will be the core battlefield of the district's elderly care talent workforce construction during the "15th Five-Year Plan". The predicted demand numbers and differences from the end of 2025 for each year are shown in the table below:

Table 5 Projected Demand for Community and Home-Based Caregivers in Xuhui District During the "15th Five-Year Plan" Period

Year	Projected Demand for Community and Home-Based Caregivers (Persons)	Difference from Actual Number at the End of 2025 (Persons)
2026	776	356
2027	791	371
2028	805	385

2029	820	400
2030	834	414

Second, the demand of elderly people in this district for community and home-based elderly care services is dominated by domestic cleaning and daily living care, with relatively less demand for medical nursing. The actual demand of elderly people in this district is highly concentrated on basic life support services. Among them, domestic cleaning services cover the widest range, being the most basic and high-frequency rigid demand for home care; daily living care and bathing assistance services follow; rehabilitation and ability training involve 21.9% of orders, mainly for daily self-care ability training, reflecting the strong demand of elderly people to maintain and restore self-care ability. In contrast, there is less demand for safety care and companionship, agency and errand services, and medical nursing services. Specific details are shown in Table 6 below:

Table 6 Community and Home-Based Care Service Needs of the Elderly in Xuhui District

Service Category	Order Proportion Involved	Main Service Content
Domestic Cleaning Services	37.8%	Daily cleaning, kitchen and bathroom cleaning, laundry and drying, making beds
Daily Living Care and Bathing Assistance	24.5%	Warm water sponge bath, bathing, head and face cleaning, assistance with eating and excretion, morning and evening care
Rehabilitation and Ability Training	21.9%	Self-care ability training, mobility with assistive devices
Meal Assistance Services	15.8%	On-site home cooking
Safety Care and Companionship	8.1%	Safety care, 24-hour care, night care, hospital companionship, medical escort
Agency and Errand Services	1.3%	Purchasing on behalf, prescription filling on behalf, escorting/handling payment on behalf
Medical Nursing Services	0.6%	Medication administration, vital sign monitoring, pressure ulcer prevention, indwelling catheter care, etc.

Note: Data is derived from a total of 25,900 single service records of registered elderly people from representative elderly care service enterprises in this district. Since a single order can involve multiple service categories simultaneously, the sum of the proportions of each category is greater than 100%.

It is worth noting that the actual consumption of the elderly in this district is highly concentrated on domestic cleaning and daily living care (accounting for more than 60% in total), while medical nursing accounts for only 0.6%, which deviates seriously from the high willingness to pay for medical and health services in the demand survey. It can be seen that at present, there is still a large proportion of non-professional elderly care services in home-visit elderly care services

in Xuhui District, while the consumption of elderly care services that require professional medical and rehabilitation nursing skills and can meet professional disability care needs is suppressed by multiple reasons, resulting in the contradictory situation of "large demand, but very little actual consumption". At the same time, the highly educated group in Xuhui District pursues personalized care with spiritual dignity, but because existing living and medical security is relatively complete, existing home-visit elderly care services find it difficult to provide differentiated elderly care services that meet the pain points of group needs, resulting in their real demand being difficult to convert into market orders. Therefore, the training of the elderly caregiver team must jump out of the homogenized and basic service stereotype, and gradually transform into professional roles with high-quality medical nursing and rehabilitation nursing capabilities, while having humanistic empathy and dynamic adaptability to different personalized service capabilities.

2.4 The Professionalism and Stability of the Caregiver Workforce Urgent for Institutional Breakthroughs

First, the age structure of caregivers is relatively old, and the gender imbalance is prominent, containing risks of human resource disruption. As of June 2026, the proportion of female caregivers is as high as 94.48%, and the average age of all staff has reached 52 years, among which personnel aged 50–59 account for as high as 66.90%, and there are only 139 young caregivers under 40 years old in the whole district. "Younger elderly caring for older elderly" has become an unavoidable objective reality at the grassroots front line. In the next five years, a large number of the main workforce will enter the pre-retirement stage. If there is no replenishment of young caregivers, the risk of talent workforce disruption is relatively large.

Second, the overall education level is relatively low, the skill structure exhibits typical bottom-heavy sedimentation characteristics, and high-level professional talents are scarce. From the perspective of educational structure, junior high school and senior high school education are the mainstream, and management and technical reserve talents with college degrees or above account for only 7.21%. Although the overall certificate holding rate of caregivers in the whole district has reached 83.9%, most of them are concentrated in primary skills and basic medical care certificates, and care talents with senior and technician levels are extremely rare. This lag in education and professional skills causes the existing workforce to face significant technical barriers and panic over professional competency.

Third, salary levels are relatively low, and the contradiction between piece-rate and working hours triggers practical labor pressure. Wage monitoring of caregivers in early 2026 shows that the average monthly salary of elderly caregivers in this district is about RMB 5,300, which lacks sufficient attractiveness. In addition, national survey data from Fushoukang shows

that more than 60% of practitioners are in a state of severe overtime work and overload operation. Long-term overtime and high-pressure work, lack of legal protection for overtime pay, coupled with low social respect for the industry, are important reasons leading to the aging of the team, difficulties in recruiting young talent, and high turnover rates in the industry.

Fourth, high living costs in the central urban area severely squeeze the retention space for migrant caregivers. As a core urban area, Xuhui District's high rent and living costs are a hard threshold that hinders migrant caregivers from taking root. High living costs directly offset the sense of gain from remuneration, leading to an annual turnover rate of caregivers hovering at a high level of 50%–60%.

Table 7 Current Status and Structure of Elderly Caregiver Team in Xuhui District

Structural Dimension	Category	Number of Persons	Proportion
Overall	Total Caregivers	2,553 Persons	100.00%
Gender	Male	141 Persons	5.52%
	Female	2,412 Persons	94.48%
Age	29 Years Old and Below	43 Persons	1.68%
	30–39 Years Old	96 Persons	3.76%
	40–49 Years Old	450 Persons	17.63%
	50–59 Years Old	1,710 Persons	66.98%
	60 Years Old and Above	254 Persons	9.95%
Education Background	Primary School and Below	248 Persons	9.71%
	Junior High School, Senior High School	2,121 Persons	83.07%
	Junior College and Above	184 Persons	7.21%
Skill	Primary & Medical Care Certificate	1,188 Persons	46.53%
	Intermediate	721 Persons	28.24%
	Senior	212 Persons	8.30%
	Technician	23 Persons	0.90%
	Uncertified	411 Persons	16.10%

Note: The number of persons and proportions across dimensions of gender, age, education, and skills are derived from the statistics of elderly caregivers across the district by the Civil Affairs Department of Xuhui District in June 2026, and the average monthly salary is derived from the estimated wage monitoring figures at the beginning of the year.

III. Countermeasure Suggestions on the Construction of the Elderly Caregiver Team in Xuhui District

3.1 Activate New Momentum of Silver Economy and Categorically Guide Multi-Level Elderly Care Consumption Needs

Our district should, on the premise of firmly maintaining the bottom line of inclusive basic elderly care services, deeply embed the caregiver workforce construction into the overall framework of cultivating the silver economy, and accurately guide multi-level elderly care consumption demands by category.

First, firmly maintain the bottom line of basic inclusive safety nets and precisely tap the quality elderly care market for highly educated and high-net-worth groups. In response to the demands of highly educated and high-net-worth seniors in the district who value quality, dignity, life autonomy, and spiritual values, policy guidance should shift to a dual-track parallel model of "bottom-line guarantee + quality incentive", encouraging the docking of commercial pension insurance, the orderly supply of high-quality care and customized health management products, guiding senior groups to pay reasonable prices for high-quality services, promoting the formation of a sustainable development pattern of "higher quality, better price", and precisely releasing differentiated consumption potential. **Second, promote the transformation of functional roles into composite types and break trust and privacy barriers with digital and intelligent means.** Promote caregivers to evolve into composite talents who combine care, psychology, and intelligent design. At the same time, explore using AI-driven intelligent customer service and demand-matching systems to assist caregivers in customizing service plans for seniors, eliminating elderly resistance and privacy barriers.

3.2 Adhere to Both Scale Increase and Quality Improvement, and Comprehensively Enhance Professional Quality of Fine Care for Disability and Quality Services

Facing the dual challenges of the rigid increase in the scale of advanced-age severe disability and high-quality personalized care, the construction of the elderly caregiver workforce in the whole district must implement a skill quality enhancement strategy that "attaches importance to quantity, but even more to quality", and vigorously introduce digital and intelligent technological means to empower talent supply.

First, reshape the skills training system and comprehensively enhance fine care capabilities. Training must break through the scope of traditional living cleaning, reconstruct the skill certification system for elderly care talents across the district, and comprehensively extend training to professional fields such as cognitive impairment, coexistence of chronic diseases, and psychological counseling; formulate differentiated skill subsidies to incentivize level leaping for

primary caregivers. **Second, focus on the demand characteristics of highly educated elderly groups and precisely create high-quality products and services.** For the highly educated senior group in Xuhui District, policies should abandon low-price gimmicks, adhere to high quality with fair prices, respect seniors' decision-making autonomy and privacy boundaries, and simultaneously build knowledge inheritance and intergenerational consultation platforms to unleash their residual intellectual potential. In implementation, set up professional windows in sub-district/town elderly care service centers to interface with commercial insurance, medical institutions, and senior universities, forming a coordinated mechanism of insurance and services. **Third, innovatively introduce AI digital and intelligent education technology to reshape the training ecosystem.** By introducing increasingly mature AI intelligent care simulation systems and 3D emergency response training platforms on the market, caregivers can conduct practical drills in highly realistic scenarios. The AI system can perform quantitative evaluation and personalized error correction on care procedures, thereby significantly shortening the training cycle. **Fourth, promote smart senior care equipment and comprehensively reshape front-line care efficiency through "Digital Intelligence – Embodied Coupling".** It is suggested to increase smart products and assistive devices in elderly care institutions and community-embedded facilities across the district, gradually freeing caregivers from heavy physical labor and improving efficiency.

Fifth, actively connect with municipal policy dividends and promote institutional breakthroughs in elderly care human resources construction. Shanghai Municipality has issued the *Several Measures on Supporting Elderly Care Service Institutions to Strengthen Human Resources Construction and Serve the Development of the Elderly Caregiver Team in the Entire Industry*. Xuhui District should take the initiative to undertake the implementation of municipal policies, focusing on forming breakthroughs in the following links: First, rely on the talent service module of the municipal elderly care service platform to smooth the supply-demand docking channels among elderly care service institutions, colleges/universities, and social job seekers; second, promote the inclusion of outstanding caregivers in the jurisdiction into the scope of district-level household registration (Hukou) policies, allowing eligible high-skilled caregivers to enjoy relevant talent introduction policies of this municipality; third, encourage elderly care service institutions in the district to establish school-enterprise cooperation alliances with institutions offering majors related to elderly care services, actively apply for municipal youth internship bases, and enjoy corresponding subsidies according to regulations; fourth, support eligible elderly care service institutions in applying for filing as social training evaluation organizations to carry out vocational skill level recognition services in the field of elderly care for the public; fifth, rely on housing security clauses in municipal policies to actively interface with affordable rental housing supplies such as the "New Era Builders and Managers Home", establish a regular contact mechanism at the district level, and provide convenience for the collective renting of elderly care service institutions. By actively connecting with municipal policy resources,

municipal institutional advantages will be transformed into regional development momentum, providing solid institutional guarantee for the high-quality development of the district's elderly caregiver team.

3.3 Connect the "Institution-Community-Home" Service Chain and Promote the Precise Leaning of Professional Care Resources toward Community and Home Care

In response to high-frequency and fine-grained rigid care demands in community and home scenarios, Xuhui District urgently needs to break the isolated pattern of supply entity scenarios and build an integrated smart service network with elderly care institutions as professional support radiating outward to community and home settings.

First, strengthen the professional hub and technology radiation functions of elderly care institutions to achieve downward extension of high-quality care resources. It should be clarified in policies that public and publicly built privately operated elderly care institutions are positioned as talent reserve pools for their respective sub-districts and towns, promoting the extension and radiation of institutional resources to surrounding community and home scenarios. On this basis, family elderly care beds should be vigorously developed to provide a solid professional base for home care. **Second, focus on rigid demand and precisely configure community and home labor resources.** Workforce construction must closely follow rigid demand service projects, change the tendency of blindly promoting advanced clinical medical nursing, and targetedly recruit and allocate community and home care forces for high-frequency scenarios, ensuring that grassroots basic life care work orders receive prompt response and high-quality delivery. **Third, promote the transformation of traditional day care centers toward differentiated functions, creating a platform hub for diversified cultivation and integrated services.** In view of issues such as single functions, fixed personnel, and narrow service radiation of traditional day care centers, professional day care institutions should be built. According to the embedded surrounding business formats (such as elderly care institutions, medical institutions, community activity centers, etc.), flexibly position differentiated functions such as cognitive impairment intervention, professional care, intensive rehabilitation physiotherapy, medical-elderly integration, or meal assistance socialization and community mobilization. This allows day care institutions to become not only platform hubs for community and home services, but also realistic carriers for diversified caregiver cultivation. Caregivers receive targeted training according to the service positioning of their center, thereby possessing the agility to flexibly operate between different scenarios of institutions, communities, and homes cross-scenario collaborative capabilities of daily life services. **Fourth, overlay multi-functional composite post settings to broaden caregivers' career paths and competence boundaries. Actively overlay functional modules such as home-visit service dispatch stations and realistic aging-friendly renovation experience points within day care institutions, comprehensively**

implementing a "one post, multiple uses" mechanism for caregivers. Support caregivers in extending into high value-added service fields such as aging-friendly renovation consulting and capability assessment, effectively broadening the competence boundaries and income sources of frontline personnel. Through multi-scenario and composite post settings, provide caregivers with clear, diversified career development channels, fundamentally and effectively enhancing the industry attractiveness and overall team stability of elderly care positions.

3.4 Improve the Comprehensive Security System for Frontline Caregivers and Take Multiple Measures to Alleviate Employment Contradictions and Living Cost Pressures

First, it is suggested to draw on advanced experience from sibling districts to create the "Home of Elderly Caregivers" security project featuring the warmth of the central urban area. The district government can coordinate to specifically build the "Home of Elderly Caregivers" or low-cost, high-quality "Youth Caregiver Hostels", providing targeted rental living subsidies and unified children's education support for young core talents who have been rooted in the frontline for a long time. **Second, promote flexible technological burden reduction and organizational model innovation.** Xuhui District can attempt to promote transfer robots and AI auxiliary decision-making systems to reduce decision fatigue through risk prediction and intervention suggestions, automatically integrate care plans, and simplify administrative paperwork. At the policy and industry levels, policy support and guidance should be strengthened to promote the establishment of reasonable salary gradients and career transition mechanisms between elderly caregivers and similar domestic service practitioners and hospital nursing aides, ensuring that caregiver salaries are competitive in the central urban area; at the same time, actively carry out public publicity and positive policy guidance to improve the whole society's recognition and acceptance of reasonable elderly care fee standards and professional value. **Lastly, it is recommended to construct a career promotion and continuing education support system for young elderly caregivers, completing the closed-loop development cycle of "attracting, performing well, and retaining".** Collaborate with local colleges, universities, and vocational institutions to offer "academic enhancement + skill certification" dual-track continuing education programs for in-service young caregivers, establishing special scholarships and paid study leave, encouraging young caregivers to complete advanced studies for junior college or undergraduate degrees on the job, and directly linking academic qualifications with the promotion of the "Eight-Level Worker" ranking. At the same time, while opening up a complete growth channel from frontline care to community management and meeting the diverse needs of family members, caregiver training needs to explore establishing horizontal job transfer channels between elderly caregivers and adjacent posts such as community social workers, rehabilitation therapists, and

elderly health managers, clarifying transfer conditions and assessment procedures, allowing young caregivers to see career prospects of "changing tracks without leaving the industry".

Annexes:

Annex 1: Results of Calculation on the Number of Disabled Elderly and Elderly Caregiver Gap in Xuhui District

Annex 2: Results of Population Estimation and Prediction of the Elderly in Shanghai and Xuhui District

Annex 3: Technical Specification for the Calculation of the Number of Disabled Elderly and the Shortage of Elderly Caregivers in Xuhui District

Annex 4: Technical Specification for the Estimation and Prediction of the Elderly Population in Shanghai and Xuhui District

Annex 1: Results of Calculation on the Number of Disabled Elderly and Elderly Caregiver Gap in Xuhui District

Three population scenarios (v = low scenario / baseline scenario / high scenario) are cascaded with three disability rate scenarios, achieving a 1:1 correspondence.

Purpose of Document: Display all intermediate calculation steps year by year from 2026 to 2030, facilitating parameter adjustment and sensitivity analysis.

Sheet 1 Input Parameters: All adjustable parameters (pink for entry, yellow for default values)

Sheet 2 Annual Population: Population of 5-year age groups in Xuhui District from 2026 to 2030 (baseline + v scaling)

Sheet 3 Disabled Elderly: Number of disabled elderly calculated by three-tier borrowing method + 1:1 cascaded scenarios

Sheet 4 Care Hours: Severity decomposition + daily average hours $\rightarrow$ annual care hours

Sheet 5 Caregiver Demand: FTE conversion $\rightarrow$ reasonable target caregiver demand

Sheet 6 Gap · Static Supply: Main report scope: supply static at 2,573 persons

Sheet 7 Gap · Dynamic Supply: Sensitivity analysis: supply growth of 5%/year

Sheet 8 Report Summary

1:1 Cascade Rules: Each scenario uses matching population, disability rate, severity, and FTE parameters:

Optimistic Scenario: v low scenario dynamic multiplier + $\pi_{\text{optimistic}}$ + $\lambda=0.92 \times \mu=0.95 + \varphi=0.40 + \delta=0.18 + T=2824h$

Baseline Scenario: v baseline scenario dynamic multiplier + π_{baseline} + $\lambda=1.00 \times \mu=1.00 + \varphi=0.35 + \delta=0.22 + T=3530h$

Prudent Scenario: v high scenario dynamic multiplier + π_{prudent} + $\lambda=1.10 \times \mu=1.12 + \varphi=0.30 + \delta=0.28 + T=4236h$

Overview of Calculation Parameters

A. Population Scenario Setting (Dynamic multiplier v by year, derived from 60+)

Derivation: Historical growth rate of Shanghai 60+ from 2017 to 2022 $\rightarrow \mu=2.746\%/year, \sigma=0.932\%/year$ (sample standard deviation, $n-1=4$) $\rightarrow g_low=\mu-1.96\sigma=0.919\%, g_high=\mu+1.96\sigma=4.574\%.$

v_t formula: $v_{\{scenario, t\}} = ((1+g_{\{scenario\}})/(1+g_base))^{(t-2025)}$ — same scaling across all 5-year age groups in the same year, without back-casting.

Dynamic v Table 2026: [0.9822, 1.0178] | 2027: [0.9647, 1.0359] | 2028: [0.9476, 1.0543] | 2029: [0.9307, 1.0731] | 2030: [0.9142, 1.0921]

National Age-Specific Disability Rate Parameters $\pi_n(x)$ (%) — Data Source: Liao Shaohong, Wang Guangzhou (2021)

Age Group	Baseline	Optimistic	Prudent
60–64	4.68	3.68	5.68
65–69	6.2	5.2	7.2
70–74	8.39	7.39	9.39
75–79	10.8	9.8	11.8
80–84	20.99	19.99	21.99
85–89	28.49	27.49	29.49
≥ 90	54.4	53.4	55.4

Note: Source: Liao Shaohong, Wang Guangzhou. Disability Status and Trends of the Elderly Population in China [J]. Population Science of China, 2021, 35(1): 38–49. Methodology: Maximum entropy reconstruction of Table 5 (row margins), Table 7 (weights), and Table 8 (column margins) of the original paper + Logistic spline split into 5-year age groups. See accompanying technical specification document §4. Three scenarios adopt baseline $\pm 1pp$ envelope.

Note: C. Adjustment Coefficients $\lambda \times \mu$

$\lambda_Shanghai$	1	0.92	1.1
μ_Xuhui	1	0.95	1.12
$\lambda \times \mu$	1	0.874	1.232

Note: D. Disability Severity Distribution + Daily Average Hours — Severity Distribution Source: Zhang Yuan, Wang Wei (2021), M1 Arithmetic Mean

Severity	Baseline Proportion	Optimistic Proportion	Prudent Proportion	Baseline h/Day	Optimistic h/Day	Prudent h/Day	Description
Mild	0.5496	0.59	0.5092	0.375	0.25	0.5	BADL 1–2 items impaired
Moderate	0.3107	0.29	0.3314	3	2	4	BADL 3–4 items impaired
Severe	0.1397	0.12	0.1594	8	6	10	BADL 5–6 items or completely dependent

Note: Columns B–D severity distribution source: Zhang Yuan, Wang Wei. Prediction of Disabled Elderly Population Size and Care Time Demand [J]. Population Research, 2021, 45(6): 110–125. Methodology: For urban elderly in 4 age groups (65–74, 75–84, 85–94, 95+) × 4 time points (2020/2030/2040/2050) population divided by mild, moderate, and severe disability, perform within-group standardization → cross-group arithmetic mean → cross-year arithmetic mean, obtaining long-cycle macro baseline (54.96/31.07/13.97). Optimistic/Prudent shifted by asymmetric ±4pp/±2pp/±2pp. Daily average hours in Columns E–G are empirical settings by the estimator. See technical specification document §5.

Note: E. FTE Conversion Parameters

Parameter	Baseline	Optimistic	Prudent
φ (Assumption Ratio)	1	1	1
δ (Redundancy Coefficient)	0.22	0.18	0.28
T (Annual Standard Working Hours, h)	4480	3840	5120

F.Caregiver Supply Baseline

Existing Caregivers in 2025 (Persons) 2405 Shanghai 67,500 × Xuhui ~7% + Urban Development data verification

Annual Growth Rate · Sensitivity 0.05 Assumed annual growth rate of caregiver supply (used for Sheet 7)

Population of 5-Year Age Groups in Xuhui District from 2026 to 2030 (Baseline + v Scaling)

Formula: Three-scenario Population = Baseline Population × v; Baseline Population is derived from the Shanghai → Xuhui estimation model

【Baseline Scenario】: Elderly Population Size

Year	60+ Total	60-64	65-69	70-74	75-79	80-84	85-89	≥90
2026	35.44	5.726	9.078	9.291	4.841	2.471	2.398	1.627
2027	35.765	5.333	8.906	9.868	5.017	2.53	2.427	1.67
2028	36.082	4.994	8.661	10.452	5.239	2.544	2.459	1.717
2029	36.391	4.702	8.352	11.042	5.489	2.531	2.494	1.766
2030	36.694	4.45	7.984	11.639	5.757	2.501	2.53	1.817

Year	Number of Persons Aged 80 and Above	Proportion of Persons Aged 80+ to Elderly Aged 60+
2026	6.496	0.1833
2027	6.627	0.1853
2028	6.72	0.1863
2029	6.791	0.1866
2030	6.849	0.1867

Baseline Scenario: Number of Disabled Elderly

Year	60-64	65-69	70-74	75-79	80-84	85-89	≥90	Total Disabled Persons
2026	0.268	0.5628	0.7795	0.5228	0.5186	0.6832	0.8852	4.22
2027	0.2496	0.5522	0.8279	0.5419	0.531	0.6914	0.9086	4.303
2028	0.2337	0.537	0.8769	0.5658	0.534	0.7007	0.9341	4.382
2029	0.22	0.5178	0.9265	0.5928	0.5313	0.7105	0.9609	4.46
2030	0.2083	0.495	0.9766	0.6218	0.525	0.7209	0.9885	4.536

【Optimistic Scenario】 Elderly Population Size

Year	60+ Total	60–64	65–69	70–74	75–79	80–84	85–89	≥90
2026	35.405	5.72	9.069	9.29	4.836	2.469	2.396	1.625
2027	35.693	5.322	8.889	9.861	5.007	2.525	2.422	1.667
2028	35.974	4.979	8.635	10.437	5.223	2.536	2.452	1.712
2029	36.245	4.682	8.319	11.014	5.467	2.521	2.484	1.759
2030	36.511	4.429	7.943	11.596	5.728	2.488	2.518	1.808

【Optimistic Scenario】 Number of Disabled Elderly

Year	60–64	65–69	70–74	75–79	80–84	85–89	≥90	Total Disabled Persons
2026	0.2105	0.4716	0.6865	0.4739	0.4935	0.6586	0.8679	3.863
2027	0.1959	0.4622	0.7287	0.4907	0.5047	0.6658	0.89	3.938
2028	0.1832	0.449	0.7713	0.5119	0.507	0.674	0.9141	4.01
2029	0.1723	0.4326	0.8139	0.5358	0.5039	0.6829	0.9393	4.081
2030	0.163	0.413	0.8569	0.5614	0.4975	0.6923	0.9654	4.149

【Prudent Scenario】 Elderly Population Size

Year	60+ Total	60-64	65-69	70-74	75-79	80-84	85-89	≥90
2026	35.475	5.732	9.087	9.308	4.846	2.473	2.4	1.629
2027	35.837	5.344	8.925	9.901	5.027	2.535	2.432	1.673
2028	36.19	5.009	8.687	10.499	5.255	2.552	2.466	1.722
2029	36.537	4.72	8.385	11.102	5.511	2.541	2.504	1.773
2030	36.877	4.473	8.023	11.712	5.786	2.514	2.544	1.826

【Prudent Scenario】 Number of Disabled Elderly

Year	60-64	65-69	70-74	75-79	80-84	85-89	≥90	Total Disabled Persons
2026	0.3256	0.6543	0.874	0.5718	0.5439	0.7079	0.9023	4.58
2027	0.3035	0.6426	0.9297	0.5932	0.5575	0.7172	0.927	4.671
2028	0.2845	0.6255	0.9859	0.6201	0.5611	0.7273	0.9541	4.758
2029	0.2681	0.6037	1.042	0.6503	0.5588	0.7384	0.9823	4.844
2030	0.2541	0.5776	1.1	0.6827	0.5527	0.7501	1.012	4.929

Year-by-Year Disabled Elderly Estimation (C Scope, 10,000 Persons) — 1:1 Cascaded Scenarios

Formula: $N_d = \sum_x [\text{pop}_x(v) \times \pi_n(x, \text{Scenario}) \times \lambda \times \mu (\text{Scenario})]$

【Scenario: Optimistic】

Year	60–64	65–69	70–74	75–79	80–84	85–89	≥90	Total Disabled	60+ Population	Disability Rate %
2026	0.184	0.4122	0.6	0.4142	0.4313	0.5756	0.7586	3.376	35.405	9.535
2027	0.1712	0.404	0.6369	0.4289	0.4411	0.582	0.7779	3.442	35.693	9.643
2028	0.1601	0.3924	0.6741	0.4474	0.4431	0.589	0.7989	3.505	35.974	9.744
2029	0.1506	0.3781	0.7114	0.4683	0.4404	0.5968	0.8209	3.566	36.245	9.84
2030	0.1424	0.361	0.749	0.4906	0.4348	0.6051	0.8438	3.627	36.511	9.933

【Scenario: Baseline】

Year	60–64	65–69	70–74	75–79	80–84	85–89	≥90	Total Disabled	60+ Population	Disability Rate %
2026	0.268	0.5628	0.7795	0.5228	0.5186	0.6832	0.8852	4.22	35.44	11.908
2027	0.2496	0.5522	0.8279	0.5419	0.531	0.6914	0.9086	4.303	35.765	12.03
2028	0.2337	0.537	0.8769	0.5658	0.534	0.7007	0.9341	4.382	36.082	12.145
2029	0.22	0.5178	0.9265	0.5928	0.5313	0.7105	0.9609	4.46	36.391	12.255
2030	0.2083	0.495	0.9766	0.6218	0.525	0.7209	0.9885	4.536	36.694	12.362

【Scenario: Prudent】

Year	60–64	65–69	70–74	75–79	80–84	85–89	≥90	Total Disabled	60+ Population	Disability Rate %
2026	0.4011	0.8061	1.077	0.7045	0.6701	0.8721	0.902	5.433	35.48	15.31
2027	0.3739	0.7917	1.145	0.7308	0.6868	0.8835	0.916	5.528	35.84	15.43
2028	0.3505	0.7706	1.215	0.7639	0.6913	0.8961	0.938	5.625	36.19	15.54

Year	60–64	65–69	70–74	75–79	80–84	85–89	≥90	Total Disabled	60+ Population	Disability Rate %
2029	0.3303	0.7438	1.284	0.8012	0.6884	0.9097	0.950	5.708	36.54	15.62
2030	0.313	0.7117	1.355	0.8411	0.681	0.9242	0.971	5.797	36.88	15.72

4_Care Hours

Annual Care Hours Estimation (10,000 Hours/Year) — 1:1 Cascaded Scenarios

Formula: $H = \sum_i [N_i \times h_i \times 365]$

【Scenario: Optimistic】

Year	Disabled Elderly (10,000)	Mild Persons	Moderate Persons	Severe Persons	Mild Hours	Moderate Hours	Severe Hours	Total Hours
2026	3.863	2.279	1.12	0.4635	207.95	817.69	1,015.1	2,040.7
2027	3.938	2.323	1.142	0.4726	212.02	833.7	1,034.9	2,080.6
2028	4.01	2.366	1.163	0.4813	215.91	849.02	1,054	2,118.9
2029	4.081	2.408	1.183	0.4897	219.69	863.87	1,072.4	2,155.9
2030	4.149	2.448	1.203	0.4979	223.4	878.44	1,090.5	2,192.3

(Continued Table 2/3: Continued from the same years in the above table, displaying remaining columns)

Year	Total Required Hours	Total Required Persons	Institution Static Supply Share	Institution Hours	Institution Required Persons	Institution Supply-Demand Static Gap	Community & Home Static Share	Home Care Required Hours	Home Service Required Caregivers	Community Supply-Demand Static Gap
2026	20,407,045	4,555.1	0.15	3,061,057	683.27	-453.73	0.1	2,040,704	455.51	35.514
2027	20,806,499	4,644.3	0.15	3,120,975	696.65	-440.35	0.1	2,080,650	464.43	44.431
2028	21,188,941	4,729.7	0.15	3,178,341	709.45	-427.55	0.1	2,118,894	472.97	52.967
2029	21,559,406	4,812.4	0.15	3,233,911	721.86	-415.14	0.1	2,155,941	481.24	61.237
2030	21,923,246	4,893.6	0.15	3,288,487	734.04	-402.96	0.1	2,192,325	489.36	69.358

(Continued Table 3/3: Continued from the same years in the above table, displaying remaining columns)

Year	Institution Dynamic Supply Share	Institution Dynamic Service Required Hours	Institution Dynamic Required Persons	Institution Supply-Demand Dynamic Gap	Dynamic Supply Share	Community Dynamic Service Required Hours	Community Dynamic Required Persons	Community Supply-Demand Dynamic Gap
2026	0.1669	3,406,719	760.43	-376.57	0.1119	2,283,548	509.72	89.721
2027	0.1704	3,545,133	791.32	-345.68	0.115	2,392,747	534.1	114.1
2028	0.1734	3,675,051	820.32	-316.68	0.1183	2,506,652	559.52	139.52
2029	0.1761	3,796,050	847.33	-289.67	0.1216	2,621,624	585.18	165.18
2030	0.1783	3,907,907	872.3	-264.7	0.1249	2,738,213	611.21	191.21

【Scenario: Baseline】

Year	Disabled Elderly (10,000)	Mild Persons	Moderate Persons	Severe Persons	Mild Hours	Moderate Hours	Severe Hours	Total Hours
2026	4.22	2.319	1.311	0.5896	317.47	1,435.8	1,721.5	3,474.8
2027	4.303	2.365	1.337	0.6011	323.67	1,463.8	1,755.2	3,542.7
2028	4.382	2.408	1.362	0.6122	329.66	1,490.9	1,787.6	3,608.2
2029	4.46	2.451	1.386	0.623	335.49	1,517.3	1,819.3	3,672
2030	4.536	2.493	1.409	0.6337	341.23	1,543.3	1,850.4	3,734.9

(Continued Table 2/3: Continued from the same years in the above table, displaying remaining columns)

Year	Total Required Hours	Total Required Persons	Institution Static Supply Share	Institution Hours	Institution Caregiver Required Persons	Institution Supply-Demand Static Gap	Community & Home Share	Community Care Required Hours	Community Service Required Caregivers	Community Supply-Demand Static Gap
2026	34,748,012	7,756.3	0.15	5,212,202	1,163.4	26.438	0.1	3,474,801	775.63	355.63
2027	35,426,753	7,907.8	0.15	5,314,013	1,186.2	49.164	0.1	3,542,675	790.78	370.78
2028	36,081,509	8,053.9	0.15	5,412,226	1,208.1	71.086	0.1	3,608,151	805.39	385.39

Year	Total Required Hours	Total Required Persons	Institution Static Supply Share	Institution Hours	Institution Caregiver Required Persons	Institution Supply-Demand Static Gap	Community & Home Share	Community Care Required Hours	Community Service Required Caregivers	Community Supply-Demand Static Gap
2029	36,720,398	8,196.5	0.15	5,508,060	1,229.5	92.478	0.1	3,672,040	819.65	399.65
2030	37,348,643	8,336.8	0.15	5,602,297	1,250.5	113.51	0.1	3,734,864	833.68	413.68

(Continued Table 3/3: Continued from the same years in the above table, displaying remaining columns)

Year	Dynamic Supply Share	Institution Dynamic Service Required Hours	Institution Dynamic Required Persons	Institution Supply-Demand Dynamic Gap	Dynamic Supply Share	Community Dynamic Service Required Hours	Community Dynamic Required Persons	Community Supply-Demand Dynamic Gap
2026	0.1669	5,800,776	1,294.8	157.82	0.1119	3,888,303	867.92	447.92
2027	0.1704	6,036,218	1,347.4	210.37	0.115	4,074,077	909.39	489.39
2028	0.1734	6,258,046	1,396.9	259.89	0.1183	4,268,442	952.78	532.78
2029	0.1761	6,465,505	1,443.2	306.19	0.1216	4,465,200	996.7	576.7
2030	0.1783	6,657,546	1,486.1	349.06	0.1249	4,664,846	1,041.3	621.26

【Scenario: Prudent】

Year	Disabled Elderly (10,000)	Mild Persons	Moderate Persons	Severe Persons	Mild Hours	Moderate Hours	Severe Hours	Total Hours
2026	4.58	2.332	1.518	0.73	425.59	2,215.9	2,664.5	5,306
2027	4.671	2.378	1.548	0.7445	434.04	2,259.9	2,717.4	5,411.3
2028	4.758	2.423	1.577	0.7585	442.2	2,302.3	2,768.5	5,513
2029	4.844	2.467	1.605	0.7722	450.16	2,343.8	2,818.4	5,612.3
2030	4.929	2.51	1.633	0.7856	458.02	2,384.7	2,867.6	5,710.3

(Continued Table 2/3: Continued from the same years in the above table, displaying remaining columns)

Year	Total Required Hours	Total Required Persons	Institution Static Supply Share	Institution Hours	Institution Caregiver Required Persons	Institution Supply-Demand Static Gap	Community & Home Share	Community Care Required Hours	Community Service Required Caregivers	Community Supply-Demand Static Gap
2026	53,060,126	11,843.8	0.15	7,959,019	1,776.6	639.57	0.1	5,306,013	1,184.4	764.38
2027	54,113,035	12,078.8	0.15	8,116,955	1,811.8	674.82	0.1	5,411,303	1,207.9	787.88
2028	55,130,404	12,305.9	0.15	8,269,561	1,845.9	708.88	0.1	5,513,040	1,230.6	810.59
2029	56,123,148	12,527.5	0.15	8,418,472	1,879.1	742.12	0.1	5,612,315	1,252.7	832.75
2030	57,103,431	12,746.3	0.15	8,565,515	1,911.9	774.95	0.1	5,710,343	1,274.6	854.63

(Continued Table 3/3: Continued from the same years in the above table, displaying remaining columns)

Year	Dynamic Supply Share	Institution Dynamic Service Required Hours	Institution Dynamic Required Persons	Institution Supply-Demand Dynamic Gap	Dynamic Supply Share	Community Dynamic Service Required Hours	Community Dynamic Required Persons	Community Supply-Demand Dynamic Gap
2026	0.1669	8,857,771	1,977.2	840.18	0.1119	5,937,428	1,325.3	905.32
2027	0.1704	9,220,096	2,058.1	921.06	0.115	6,222,999	1,389.1	969.06
2028	0.1734	9,561,923	2,134.4	997.36	0.1183	6,521,927	1,455.8	1,035.8
2029	0.1761	9,881,824	2,205.8	1,068.8	0.1216	6,824,575	1,523.3	1,103.3
2030	0.1783	10,178,917	2,272.1	1,135.1	0.1249	7,132,219	1,592	1,172

Caregiver FTE Demand Estimation (Persons) — 1:1 Cascaded Scenarios — Overall Situation of Xuhui District

Formula: $W = H \times 10^4 \times \varphi \times (1 + \delta) / T$

Year	Baseline Demand	Optimistic Demand	Prudent Demand
2026	9,462.6	6,270.9	13,265
2027	9,647.5	6,393.7	13,528.3
2028	9,825.8	6,511.2	13,782.6
2029	9,999.8	6,625	14,030.8
2030	10,170.8	6,736.8	14,275.9

Main Report Scope: Caregiver Gap (Static Supply = 2,405 Persons) — 1:1 Cascaded Scenarios

Formula: $Gap_t = W_t - S$

Year	Existing Supply	Baseline Gap	Optimistic Gap	Prudent Gap
2026	2405	7,057.6	3,865.9	10,860
2027	2405	7,242.5	3,988.7	11,123.3
2028	2405	7,420.8	4,106.2	11,377.6
2029	2405	7,594.8	4,220	11,625.8
2030	2405	7,765.8	4,331.8	11,870.9

Sensitivity Analysis: Caregiver Gap (Supply 5%/Year Growth) — 1:1 Cascaded Scenarios

Formula: $S_t = S_0 \times (1+r)^{(t-2025)}$, $r=5\%$ (can be adjusted in 1_Input Parameters!B39)

Year	Dynamic Supply	Baseline Gap	Optimistic Gap	Prudent Gap
2026	2,525.2	6,937.4	3,745.7	10,739.8
2027	2,651.5	6,996	3,742.2	10,876.7
2028	2,784.1	7,041.7	3,727.1	10,998.5
2029	2,923.3	7,076.5	3,701.7	11,107.5
2030	3,069.5	7,101.4	3,667.4	11,206.4

8_Report Summary

Core Results Summary — 1:1 Cascaded Scenarios — Overall Situation of Xuhui District

1. Caregiver Demand (Reasonable Target FTE, Persons) — 1:1 Cascade

Year	Baseline	Optimistic	Prudent	Interval Expression
2026	9,462.6	6,270.9	13,265	9,463 Persons (Interval: 6,271–13,265 Persons)
2027	9,647.5	6,393.7	13,528.3	9,647 Persons (Interval: 6,394–13,528 Persons)
2028	9,825.8	6,511.2	13,782.6	9,826 Persons (Interval: 6,511–13,783 Persons)
2029	9,999.8	6,625	14,030.8	10,000 Persons (Interval: 6,625–14,031 Persons)
2030	10,170.8	6,736.8	14,275.9	10,171 Persons (Interval: 6,737–14,276 Persons)

2.Gap · Static Supply (Referencing 2,405 Persons) — 1:1 Cascade

Year	Baseline	Optimistic	Prudent	Interval Expression
2026	7,057.6	3,865.9	10,860	7,058 Persons (Interval: 3,866–10,860 Persons)
2027	7,242.5	3,988.7	11,123.3	7,242 Persons (Interval: 3,989–11,123 Persons)
2028	7,420.8	4,106.2	11,377.6	7,421 Persons (Interval: 4,106–11,378 Persons)
2029	7,594.8	4,220	11,625.8	7,595 Persons (Interval: 4,220–11,626 Persons)
2030	7,765.8	4,331.8	11,870.9	7,766 Persons (Interval: 4,332–11,871 Persons)

3. Gap · Dynamic Supply (5%/Year Growth) — 1:1 Cascade

Year	Baseline	Optimistic	Prudent	Interval Expression
2026	6,937.4	3,745.7	10,739.8	6,937 Persons (Interval: 3,746–10,740 Persons)
2027	6,996	3,742.2	10,876.7	6,996 Persons (Interval: 3,742–10,877 Persons)
2028	7,041.7	3,727.1	10,998.5	7,042 Persons (Interval: 3,727–10,999 Persons)
2029	7,076.5	3,701.7	11,107.5	7,076 Persons (Interval: 3,702–11,107 Persons)
2030	7,101.4	3,667.4	11,206.4	7,101 Persons (Interval: 3,667–11,206 Persons)

9_Institutional Scope Conversion

IX. Explanation of Institutional Scope Conversion Coefficients

A. Module Positioning

This worksheet is a supplementary explanation of the five-level cascade model and does not modify the calculation logic of the original model. The caregiver demand scale obtained in the fourth stage of the original model belongs to the full scope, that is, covering all service entities undertaking formal care work such as elderly care institutions, community nursing stations, and medical institutions. By introducing institutional scope conversion coefficients, this worksheet converts the full-scope demand into the actual workforce construction demand scale under the single entity scope of elderly care institutions, facilitating policymakers to identify the scope differences between the calculation results of this study and actual training plans of elderly care institutions.

B. Distribution and Evolution of National Nursing Service Market Entities (2022–2030)

Year	Hospital Nursing Service Proportion	Community Nursing Service Proportion	Elderly Care Institution Nursing Service Proportion	Community Home-Based Care Proportion	Description
2022	0.65	0.2	0.15	0.1	Base Period
2026	0.5588	0.2743	0.1669	0.1119	Linear extrapolation based on annual average growth rate of each category
2027	0.5347	0.2949	0.1704	0.115	Linear extrapolation based on annual average growth rate of each category
2028	0.5103	0.3163	0.1734	0.1183	Linear extrapolation based on annual average growth rate of each category
2029	0.4856	0.3383	0.1761	0.1216	Linear extrapolation based on annual average growth rate of each category
2030	0.4609	0.3608	0.1783	0.1249	Linear extrapolation based on annual average growth rate of each category

Data base period: 2022 nursing service market structure (hospitals 65%, community 20%, elderly care institutions 15%, non-profit institutions 8%). Growth rate: hospitals 5%/year, community 18%/year, elderly care institutions 12%/year (based on user-provided data). Extrapolation method: The base period scale of each category is linearly extrapolated by the average annual growth rate and normalized to obtain the proportion of each year. Non-profit institutions, as a cross-category attribute, account for about 8% of the total market, mainly distributed in elderly care institutions and community nursing service fields, and are not listed separately.

Gap Details See Below

【Scenario: Baseline】

Year	Disabled Elderly (10,000)	Mild Persons	Moderate Persons	Severe Persons	Mild Hours	Moderate Hours	Severe Hours	Total Hours
2026	4.22	2.319	1.311	0.5896	317.47	1,435.8	1,721.5	3,474.8
2027	4.303	2.365	1.337	0.6011	323.67	1,463.8	1,755.2	3,542.7
2028	4.382	2.408	1.362	0.6122	329.66	1,490.9	1,787.6	3,608.2
2029	4.46	2.451	1.386	0.623	335.49	1,517.3	1,819.3	3,672
2030	4.536	2.493	1.409	0.6337	341.23	1,543.3	1,850.4	3,734.9

Year	Total Required Hours	Total Required Persons	Institution Static Supply Share	Institution Hours	Institution Caregiver Required Persons	Institution Supply-Demand Static Gap	Community & Home Share	Community Care Required Hours	Community Service Required Caregivers	Community Supply-Demand Static Gap
2026	34,748,012	7,756.3	0.15	5,212,202	1,163.4	26.438	0.1	3,474,801	775.63	355.63
2027	35,426,753	7,907.8	0.15	5,314,013	1,186.2	49.164	0.1	3,542,675	790.78	370.78

Year	Total Required Hours	Total Required Persons	Institution Static Supply Share	Institution Hours	Institution Caregiver Required Persons	Institution Supply-Demand Static Gap	Community & Home Share	Community Care Required Hours	Community Service Required Caregivers	Community Supply-Demand Static Gap
2028	36,081,509	8,053.9	0.15	5,412,226	1,208.1	71.086	0.1	3,608,151	805.39	385.39
2029	36,720,398	8,196.5	0.15	5,508,060	1,229.5	92.478	0.1	3,672,040	819.65	399.65
2030	37,348,643	8,336.8	0.15	5,602,297	1,250.5	113.51	0.1	3,734,864	833.68	413.68
	Year	Dynamic Supply Share	Institution Dynamic Service Required Hours	Institution Dynamic Required Persons	Institution Supply-Demand Dynamic Gap	Dynamic Supply Share	Community Dynamic Service Required Hours	Community Dynamic Required Persons	Community Supply-Demand Dynamic Gap	
	2026	0.1669	5,800,776	1,294.8	157.82	0.1119	3,888,303	867.92	447.92	
	2027	0.1704	6,036,218	1,347.4	210.37	0.115	4,074,077	909.39	489.39	
	2028	0.1734	6,258,046	1,396.9	259.89	0.1183	4,268,442	952.78	532.78	
	2029	0.1761	6,465,505	1,443.2	306.19	0.1216	4,465,200	996.7	576.7	
	2030	0.1783	6,657,546	1,486.1	349.06	0.1249	4,664,846	1,041.3	621.26	

【Scenario: Optimistic】

Year	Disabled Elderly (10,000)	Mild Persons	Moderate Persons	Severe Persons	Mild Hours	Moderate Hours	Severe Hours	Total Hours
2026	3.863	2.279	1.12	0.4635	207.95	817.69	1,015.1	2,040.7
2027	3.938	2.323	1.142	0.4726	212.02	833.7	1,034.9	2,080.6
2028	4.01	2.366	1.163	0.4813	215.91	849.02	1,054	2,118.9
2029	4.081	2.408	1.183	0.4897	219.69	863.87	1,072.4	2,155.9
2030	4.149	2.448	1.203	0.4979	223.4	878.44	1,090.5	2,192.3

Year	Total Required Hours	Total Required Persons	Institution Static Supply Share	Institution Hours	Institution Required Persons	Institution Supply-Demand Static Gap	Community & Home Static Share	Home Care Required Hours	Home Service Required Caregivers	Community Supply-Demand Static Gap
2026	20,407,045	4,555.1	0.15	3,061,057	683.27	-453.73	0.1	2,040,704	455.51	35.514
2027	20,806,499	4,644.3	0.15	3,120,975	696.65	-440.35	0.1	2,080,650	464.43	44.431
2028	21,188,941	4,729.7	0.15	3,178,341	709.45	-427.55	0.1	2,118,894	472.97	52.967
2029	21,559,406	4,812.4	0.15	3,233,911	721.86	-415.14	0.1	2,155,941	481.24	61.237
2030	21,923,246	4,893.6	0.15	3,288,487	734.04	-402.96	0.1	2,192,325	489.36	69.358

Year	Institution Dynamic Supply Share	Institution Dynamic Service Required Hours	Institution Dynamic Required Persons	Institution Supply-Demand Dynamic Gap	Dynamic Supply Share	Community Dynamic Service Required Hours	Community Dynamic Required Persons	Community Supply-Demand Dynamic Gap
2026	0.1669	3,406,719	760.43	-376.57	0.1119	2,283,548	509.72	89.721
2027	0.1704	3,545,133	791.32	-345.68	0.115	2,392,747	534.1	114.1
2028	0.1734	3,675,051	820.32	-316.68	0.1183	2,506,652	559.52	139.52
2029	0.1761	3,796,050	847.33	-289.67	0.1216	2,621,624	585.18	165.18
2030	0.1783	3,907,907	872.3	-264.7	0.1249	2,738,213	611.21	191.21

【Scenario: Prudent】

Year	Disabled Elderly (10,000)	Mild Persons	Moderate Persons	Severe Persons	Mild Hours	Moderate Hours	Severe Hours	Total Hours
2026	4.58	2.332	1.518	0.73	425.59	2,215.9	2,664.5	5,306
2027	4.671	2.378	1.548	0.7445	434.04	2,259.9	2,717.4	5,411.3
2028	4.758	2.423	1.577	0.7585	442.2	2,302.3	2,768.5	5,513
2029	4.844	2.467	1.605	0.7722	450.16	2,343.8	2,818.4	5,612.3
2030	4.929	2.51	1.633	0.7856	458.02	2,384.7	2,867.6	5,710.3

Year	Total Required Hours	Total Required Persons	Institution Static Supply Share	Institution Hours	Institution Caregiver Required Persons	Institution Supply-Demand Static Gap	Community & Home Share	Community Care Required Hours	Community Service Required Caregivers	Community Supply-Demand Static Gap
2026	53,060,126	11,843.8	0.15	7,959,019	1,776.6	639.57	0.1	5,306,013	1,184.4	764.38
2027	54,113,035	12,078.8	0.15	8,116,955	1,811.8	674.82	0.1	5,411,303	1,207.9	787.88
2028	55,130,404	12,305.9	0.15	8,269,561	1,845.9	708.88	0.1	5,513,040	1,230.6	810.59
2029	56,123,148	12,527.5	0.15	8,418,472	1,879.1	742.12	0.1	5,612,315	1,252.7	832.75
2030	57,103,431	12,746.3	0.15	8,565,515	1,911.9	774.95	0.1	5,710,343	1,274.6	854.63

Year	Dynamic Supply Share	Institution Dynamic Service Required Hours	Institution Dynamic Required Persons	Institution Supply-Demand Dynamic Gap	Dynamic Supply Share	Community Dynamic Service Required Hours	Community Dynamic Required Persons	Community Supply-Demand Dynamic Gap
2026	0.1669	8,857,771	1,977.2	840.18	0.1119	5,937,428	1,325.3	905.32
2027	0.1704	9,220,096	2,058.1	921.06	0.115	6,222,999	1,389.1	969.06
2028	0.1734	9,561,923	2,134.4	997.36	0.1183	6,521,927	1,455.8	1,035.8
2029	0.1761	9,881,824	2,205.8	1,068.8	0.1216	6,824,575	1,523.3	1,103.3
2030	0.1783	10,178,917	2,272.1	1,135.1	0.1249	7,132,219	1,592	1,172

Annex 2 - Results of Population Estimation and Prediction of the Elderly in Shanghai and Xuhui District

I. Research Objectives

Based on paired elderly population data (bulletin / yearbook) of Shanghai and Xuhui District from 2019 to 2025, construct the estimation model $Xuhui \equiv f(Shanghai)$;
Back-cast / predict the 9-threshold structure of Xuhui District under the same scope from 2017 to 2030, and subdivide it into 7 five-year age groups.

II. Data Sources

Data 1: Total population and 60+ / 65+ / 80+ (bulletin) of Shanghai and Xuhui District from 2019 to 2025; Data 2: Registered elderly population divided into 9 thresholds in Shanghai from 2017 to 2025.
Data 3: Birth rate / death rate of Xuhui District (reference). This version fully adopts the scope of bulletins/yearbooks.

III. Econometric Methods

Step 1: Calibration Regression (2019–2025, n=7):

60+ Share: $Y_{_xh} = 16.9369 + 0.5262 \cdot Y_{_sh}$ ($R^2=0.919$)

65+ Share: $Y_{_xh} = 1.0438 + 0.9737 \cdot Y_{_sh}$ ($R^2=0.999$)

80+/Total Share: $Y_{_xh} = 1.3815 + 0.9106 \cdot Y_{_sh}$ ($R^2=0.803$)

★80+/60+ Proportion (Nonlinear Log-Log): $Y_{_xh} = 1.6946 \cdot Y_{_sh}^{0.8647}$ (Elasticity=0.865, $R^2=0.985$)

【Method Change Description】 The original version used OLS linear regression for 80+/60+; this version switches to a log-log nonlinear form. An elasticity of 0.865 describes the non-proportional relationship of the advanced-age structure between Xuhui and Shanghai. As a traditional central urban area, Xuhui's advanced-age proportion ($R(80) \approx 1.14\text{--}1.17$) is higher than Shanghai, but an elasticity < 1 indicates that its relative growth rate tends to converge.

Step 2: R(a) Age-Ratio Function:

$\log R(a)$ with respect to age performs a quadratic polynomial precisely passing through three points $a=60/65/80$, smoothly interpolating the remaining 6 thresholds.

Step 3: Prediction Strategy:

Abandon simple linear extrapolation of proportions for each threshold. Changed to:

- (1) Shanghai 60+/65+ proportions adopt saturated exponential convergence extrapolation (upper limit 43.0/36.0), suppressing short-term inflated proportions caused by the Great Famine compensatory cohort;
- (2) Shanghai 80+ proportion fits a real V-shape: COVID-19 excess deaths in 2022–2023 caused a trough, starting with a recovery slope (+0.192/year) in 2023–2025, attenuating with a 2-year time constant to a structural trend (+0.019/year, Medium Scenario). Restores post-pandemic advanced aging momentum.
- (3) The remaining thresholds are linearly extrapolated. Input the new-scope Shanghai proportion into Step 1 regression + R(a) to obtain Xuhui distribution; subtract adjacent thresholds to obtain the number of persons in each five-year interval group.

IV. Uncertainty

$\pm 1\sigma$ Band (Sheet 10–13): Step 1 calibration RMSE + Step 3 extrapolation residual synthesized via error propagation.

v Three Scenarios (Sheet 14–17): Constructed based on Shanghai 2017–2022 60+ growth rate $\mu=2.746\%$ / $\sigma=0.932\%$, uniformly applied to all thresholds (avoiding back-casting of five-year groups).

V. Core Results

Xuhui 80+/60+ continues upward from 18.0% (observed) in 2025 to 18.67% (Medium Scenario) in 2030, representing the continuation of real recovery momentum with growth gradually slowing year by year, correcting the counter-intuitive decline in the original version (18.07% $\rightarrow$ 17.02%). The 60–64 group continues on a genuine downward trajectory (the pass-through effect of the Great Famine cohort).

VI. Limitations

The recovery slope is based only on the three years of 2023–2025; the saturation upper limit L, V-shaped attenuation constant, and elastic extrapolation are all judgment-based parameters;

Fundamental improvement requires a cohort-component method under age-specific mortality rates. This result is a scenario estimate rather than an exact forecast.

Step 1: Shanghai → Xuhui District Calibration Regression (2019–2025, n=7, Bulletin Data)

Indicator	Regression Equation	Parameter 1	Parameter 2	R²	RMSE	Form
60+ Share of Total Population (%)	$Y=\alpha+\beta\cdot Y_{sh}$	$\alpha=16.9369$	$\beta=0.5262$	0.9192	0.128	Linear OLS
65+ Share of Total Population (%)	$Y=\alpha+\beta\cdot Y_{sh}$	$\alpha=1.0438$	$\beta=0.9737$	0.9985	0.0685	Linear OLS
80+ Share of Total Population (%)	$Y=\alpha+\beta\cdot Y_{sh}$	$\alpha=1.3815$	$\beta=0.9106$	0.8027	0.0477	Linear OLS
80+ Share of 60+ (%)	$Y=k\cdot Y_{sh}^e$	$k=1.6946$	$e=0.8647$	0.9847	0.0569	★Log-Log Nonlinear

- The ratio of elderly people aged 80+ to elderly people aged 60+ switched to a log-log nonlinear form (elasticity 0.865): portrays the non-proportional relationship between Xuhui's advanced aging structure and Shanghai. Elasticity < 1 indicates that historically the relative growth rate of the elderly population aged 80+ in Xuhui has been slower than Shanghai (a high base tending toward saturation).
- The regression fit for the number of elderly people aged 65+ is optimal ($R^2=0.999$); $R^2=0.803$ for the number of elderly people aged 80+/total is relatively low, mainly because COVID-19 excess deaths in 2022–2023 caused abnormal fluctuations in the 80+ population in both places.

Step 1 Regression — Fitted Values vs. Observed Values (2019–2025)

Year	Shanghai 80+/60+	Xuhui 80+/60+ (Observed)	Xuhui (Log-Log Predicted)	Residual	Shanghai 60+ Share	Xuhui 60+ (Observed)	Xuhui (Linear Predicted)
2019	15.823	18.365	18.454	-0.0893	35.218	35.373	35.467
2020	15.47	18.19	18.098	0.0921	36.093	35.923	35.927
2021	15.47	18.08	18.098	-0.0177	36.261	36.082	36.016
2022	15.018	17.615	17.64	-0.0255	36.783	36.45	36.291
2023	14.372	16.939	16.982	-0.0424	37.385	36.706	36.607
2024	14.889	17.553	17.508	0.0444	37.557	36.728	36.698
2025	15.326	17.991	17.952	0.0389	37.564	36.446	36.701

R(a) = Xuhui Proportion / Shanghai Proportion — Age-Ratio Function (log quadratic polynomial, calibrated at a=60, 65, 80)

Year	R(60)	R(65)	R(70)	R(75)	R(80)	R(85)	R(90)	R(95)	R(100)
2019	1.004	1.019	1.049	1.098	1.166	1.257	1.377	1.531	1.729
2020	0.9953	1.011	1.044	1.096	1.17	1.271	1.404	1.577	1.801
2021	0.9951	1.011	1.043	1.093	1.163	1.257	1.38	1.538	1.742
2022	0.9909	1.011	1.046	1.095	1.162	1.249	1.361	1.501	1.677
2023	0.9818	1.012	1.051	1.099	1.157	1.227	1.311	1.411	1.529
2024	0.9779	1.012	1.052	1.099	1.153	1.216	1.288	1.371	1.466
2025	0.9702	1.006	1.045	1.089	1.139	1.194	1.256	1.325	1.401
2026	0.9731	1.008	1.046	1.089	1.136	1.189	1.246	1.311	1.381
2027	0.9695	1.007	1.046	1.087	1.13	1.175	1.222	1.271	1.322
2028	0.9662	1.006	1.046	1.086	1.125	1.164	1.201	1.238	1.273
2029	0.9631	1.006	1.047	1.085	1.121	1.154	1.183	1.209	1.231
2030	0.9603	1.006	1.047	1.085	1.118	1.146	1.168	1.184	1.194

Observe that R(a) increases as a rises ($R(60) \approx 0.97\text{--}1.0$, $R(80) \approx 1.14\text{--}1.17$), reflecting the internal structure of Xuhui's aging population being "older". Green indicates observed years.

Registered Elderly Population in Shanghai (2017–2025 Bulletin/Yearbook Observation · 2026–2030 New Model Extrapolation)

Year	Total Pop (10,000)	60+ Pop	65+ Pop	70+ Pop	75+ Pop	80+ Pop	85+ Pop	90+ Pop	95+ Pop	100+ Pop
2017	1,456.3	483.6	317.67	197.71	125.5	80.58	38.35	12	2.29	0.216
2018	1,463.6	503.28	336.9	208.25	127.8	81.67	41.43	13.26	2.56	0.252
2019	1,471.2	518.12	361.66	220.71	132.15	81.98	43.88	14.14	2.83	0.273
2020	1,478.1	533.49	382.44	233.47	136.03	82.53	45.91	15.15	3.14	0.308
2021	1,495.3	542.22	402.37	247.76	142.53	83.88	46.69	16	3.39	0.351
2022	1,505.2	553.66	424.4	263.17	148.94	83.15	45.87	16.8	3.48	0.353
2023	1,519.5	568.05	437.92	276.21	152.88	81.64	49.545	18.011	3.854	0.397
2024	1,538	577.62	452.14	291.6	158.7	86	51.497	19.117	4.141	0.43
2025	1,555.7	584.38	466.5	307.12	164.53	89.56	53.455	20.234	4.431	0.463
2026	1,560.1	591.15	478.63	320.18	169.01	92.81	54.975	21.19	4.687	0.493
2027	1,572.5	600.69	492.07	335.02	174.4	95.495	56.791	22.265	4.969	0.526
2028	1,584.9	610.02	504.55	350.05	179.86	97.557	58.629	23.354	5.255	0.559
2029	1,597.3	619.14	516.16	365.27	185.38	99.239	60.488	24.457	5.545	0.592
2030	1,609.7	628.06	527	380.69	190.96	100.69	62.369	25.574	5.839	0.627

(Continued Table 2/2: Continued from the same years in the above table, displaying remaining columns)

Year	60+ Share %	80+/60+ %	Nature of Data
2017	33.21	16.66	Observed (Bulletin/Yearbook)
2018	34.39	16.23	Observed (Bulletin/Yearbook)
2019	35.22	15.82	Observed (Bulletin/Yearbook)
2020	36.09	15.47	Observed (Bulletin/Yearbook)
2021	36.26	15.47	Observed (Bulletin/Yearbook)
2022	36.78	15.02	Observed (Bulletin/Yearbook)
2023	37.38	14.37	Observed (Bulletin/Yearbook)
2024	37.56	14.89	Observed (Bulletin/Yearbook)
2025	37.56	15.33	Observed (Bulletin/Yearbook)
2026	37.89	15.7	New Model Extrapolation
2027	38.2	15.9	New Model Extrapolation
2028	38.49	15.99	New Model Extrapolation
2029	38.76	16.03	New Model Extrapolation
2030	39.02	16.03	New Model Extrapolation

Estimation of Registered Elderly Population in Xuhui District (2019–2025, 60+ / 65+ / 80+ are bulletin observations, other thresholds are back-cast by R(a))

Year	Total Pop (10,000)	60+	65+	70+	75+	80+	85+	90+	95+	100+
2019	92.67	32.78	23.21	14.59	9.137	6.02	3.475	1.226	0.273	0.03
2020	93.2	33.48	24.37	15.362	9.399	6.09	3.679	1.341	0.312	0.035
2021	94.12	33.96	25.6	16.263	9.803	6.14	3.694	1.39	0.328	0.038
2022	94.54	34.46	26.96	17.285	10.247	6.07	3.599	1.436	0.328	0.037
2023	94.89	34.83	27.68	18.126	10.491	5.9	3.798	1.475	0.34	0.038
2024	95.24	34.98	28.33	18.993	10.797	6.14	3.877	1.525	0.352	0.039
2025	95.32	34.74	28.74	19.666	10.982	6.25	3.911	1.557	0.36	0.04

(Continued Table 2/2: Continued from the same years in the above table, displaying remaining columns)

Year	80+ / 60+ %
2019	18.36
2020	18.19
2021	18.08
2022	17.61
2023	16.94
2024	17.55
2025	17.99

60+ / 65+ / 80+ are direct observations from bulletins; 70+ / 75+ / 85+ / 90+ / 95+ / 100+ are back-cast by the R(a) age-ratio function. Xuhui 80+ / 60+ exhibits a true V-shape: 18.4% in 2019 → trough 16.9% in 2023 → 18.0% in 2025.

Step 3: Cumulative Distribution of 9 Thresholds in Xuhui District 2026–2030 (Medium Scenario Baseline)

Year	Total Pop (10,000)	60+	65+	70+	75+	80+	85+	90+	95+	100+
2026	96.11	35.44	29.714	20.636	11.337	6.496	4.025	1.627	0.378	0.042
2027	96.57	35.765	30.432	21.525	11.644	6.627	4.097	1.67	0.388	0.043
2028	97.03	36.082	31.088	22.427	11.959	6.72	4.176	1.717	0.398	0.044
2029	97.48	36.391	31.69	23.338	12.28	6.791	4.26	1.766	0.409	0.044
2030	97.94	36.694	32.243	24.26	12.606	6.849	4.348	1.817	0.421	0.046

(Continued Table 2/2: Continued from the same years in the above table, displaying remaining columns)

Year	80+/60+ %	Prediction Method
2026	18.33	Nonlinear regression + R(a) + V-shaped recovery
2027	18.53	Nonlinear regression + R(a) + V-shaped recovery
2028	18.63	Nonlinear regression + R(a) + V-shaped recovery
2029	18.66	Nonlinear regression + R(a) + V-shaped recovery
2030	18.67	Nonlinear regression + R(a) + V-shaped recovery

Step 3: Five-Year Age Group Population and Proportion of 60+ in Xuhui District 2026–2030 (Medium Scenario)

Year	60+ Total (10,000)	60–64 (10,000)	65–69 (10,000)	70–74 (10,000)	75–79 (10,000)	80–84 (10,000)	85–89 (10,000)	≥90 (10,000)	60–64 Share of 60+ %	65–69 Share of 60+ %
2026	35.44	5.726	9.078	9.299	4.841	2.471	2.398	1.627	16.16	25.62
2027	35.765	5.333	8.906	9.881	5.017	2.53	2.427	1.67	14.91	24.9
2028	36.082	4.994	8.661	10.467	5.239	2.544	2.459	1.717	13.84	24
2029	36.391	4.702	8.352	11.058	5.489	2.531	2.494	1.766	12.92	22.95
2030	36.694	4.45	7.984	11.653	5.757	2.501	2.53	1.817	12.13	21.76

(Continued Table 2/2: Continued from the same years in the above table, displaying remaining columns)

Year	70–74 Share of 60+ %	75–79 Share of 60+ %	80–84 Share of 60+ %	85–89 Share of 60+ %	≥90 Share of 60+ %
2026	26.24	13.66	6.97	6.77	4.59
2027	27.63	14.03	7.07	6.79	4.67
2028	29.01	14.52	7.05	6.82	4.76
2029	30.39	15.08	6.95	6.85	4.85
2030	31.76	15.69	6.82	6.9	4.95

Key Findings:

- The proportion of 80+ among 60+ rises from 18.0% (observed) in 2025 to 18.67% in 2030, continuing the genuine recovery momentum of 2023–2025, with growth slowing year by year and stabilizing.
- The share of the 60–64 group declines moderately from 16.2% to 12.1%, following the real downward trajectory during the observation period (the pass-through effect of the Great Famine compensatory cohort).
- The 70–74 group is the fastest growing group (26.2% → 31.7%); the sum of the three groups 80–84 / 85–89 / ≥90 moves upward steadily, supporting the rise of 80+/60+.

Birth Rate and Death Rate of Population in Xuhui District (Reference)

Year	Birth Rate (‰)	Death Rate (‰)	Natural Growth Rate (‰)
2019	6.28	8.61	-2.33
2020	5.17	8.56	-3.39
2021	4.68	8.96	-4.28
2022	4.53	10.72	-6.19
2023	4.33	12.56	-8.23
2024	4.69	8.58	-3.89

The mortality rate rose significantly in 2022–2023 (10.72‰ / 12.56‰), corresponding to COVID-19 excess deaths, which is the direct cause of the V-shaped trough of the 80+ population in Xuhui.

±1σ Uncertainty Band Methodology — σ_cal ⊕ σ_ext → σ_total

Method: Step 1 calibration regression residual (σ_cal=RMSE) and Step 3 extrapolation residual (σ_ext) are synthesized via error propagation: σ_total = √(σ_cal² + σ_ext²).

Threshold	Regression RMSE (σ_cal, pp)	σ_ext (pp)	σ_total (pp)	Description
60+	0.128	0.1	0.1625	Step 1 60+ Regression
65+	0.0685	0.1	0.1212	Step 1 65+ Regression
80+	0.0477	0.1	0.1108	Step 1 80+ Regression

Blue/yellow text indicates adjustable inputs; the remaining threshold σ_share values take an approximation of 0.10pp. See Sheet 11–13 for the ±1σ band.

Xuhui District 2026–2030 Cumulative Thresholds $\pm 1\sigma$ Uncertainty Band (10,000 Persons)

Threshold	Year	Baseline (10,000)	σ _count	Lower Bound -1σ	Upper Bound $+1\sigma$	Relative %	Share %
60+	2026	35.44	0.1914	35.248	35.631	0.54	36.874
60+	2027	35.765	0.1922	35.573	35.957	0.54	37.036
60+	2028	36.082	0.1931	35.889	36.275	0.54	37.188
60+	2029	36.391	0.194	36.197	36.585	0.53	37.331
60+	2030	36.694	0.1948	36.499	36.889	0.53	37.465
65+	2026	29.714	0.1489	29.565	29.863	0.5	30.916
65+	2027	30.432	0.1505	30.281	30.582	0.49	31.513
65+	2028	31.088	0.1519	30.936	31.24	0.49	32.041
65+	2029	31.69	0.1532	31.537	31.843	0.48	32.508
65+	2030	32.243	0.1544	32.089	32.398	0.48	32.922
70+	2026	20.636	0.1504	20.486	20.786	0.73	21.471
70+	2027	21.525	0.1521	21.373	21.677	0.71	22.29
70+	2028	22.427	0.1537	22.273	22.58	0.69	23.114
70+	2029	23.338	0.1554	23.183	23.494	0.67	23.941
70+	2030	24.26	0.1572	24.102	24.417	0.65	24.77
75+	2026	11.337	0.1405	11.197	11.478	1.24	11.796
75+	2027	11.644	0.1413	11.503	11.786	1.21	12.058
75+	2028	11.959	0.1421	11.817	12.101	1.19	12.326
75+	2029	12.28	0.1429	12.137	12.423	1.16	12.598
75+	2030	12.606	0.1438	12.462	12.75	1.14	12.871
80+	2026	6.496	0.1084	6.388	6.605	1.67	6.759

Threshold	Year	Baseline (10,000)	σ _count	Lower Bound -1σ	Upper Bound $+1\sigma$	Relative %	Share %
80+	2027	6.627	0.1089	6.518	6.736	1.64	6.863
80+	2028	6.72	0.1095	6.611	6.83	1.63	6.926
80+	2029	6.791	0.11	6.681	6.901	1.62	6.967
80+	2030	6.849	0.1105	6.738	6.959	1.61	6.993
85+	2026	4.025	0.1365	3.889	4.162	3.39	4.188
85+	2027	4.097	0.1372	3.96	4.234	3.35	4.243
85+	2028	4.176	0.1378	4.039	4.314	3.3	4.304
85+	2029	4.26	0.1385	4.122	4.399	3.25	4.37
85+	2030	4.348	0.1391	4.208	4.487	3.2	4.439
90+	2026	1.627	0.136	1.491	1.763	8.36	1.693
90+	2027	1.67	0.1367	1.534	1.807	8.18	1.73
90+	2028	1.717	0.1373	1.58	1.854	8	1.77
90+	2029	1.766	0.138	1.628	1.904	7.81	1.812
90+	2030	1.817	0.1386	1.678	1.956	7.63	1.855
95+	2026	0.378	0.1359	0.242	0.514	35.92	0.394
95+	2027	0.388	0.1366	0.251	0.524	35.23	0.401
95+	2028	0.398	0.1372	0.261	0.535	34.47	0.41
95+	2029	0.409	0.1379	0.271	0.547	33.7	0.42
95+	2030	0.421	0.1385	0.282	0.559	32.94	0.429
100+	2026	0.042	0.1359	-0.094	0.178	323.9	0.044
100+	2027	0.043	0.1366	-0.094	0.179	320	0.044
100+	2028	0.044	0.1372	-0.094	0.181	315.14	0.045
100+	2029	0.044	0.1379	-0.093	0.182	309.84	0.046
100+	2030	0.046	0.1385	-0.093	0.184	304.4	0.046

Xuhui District 2026–2030 Five-Year Age Groups ±1σ Band (10,000 Persons)

5-year group $\sigma^2 \approx (\text{total}/100)^2 \cdot (\sigma^2_{\text{share}_a} + \sigma^2_{\text{share}_{(a+5)}})$. Conservatively assuming that error in proportion between adjacent thresholds is independent (in reality positively correlated), so the band is slightly wide.

5-Year Group	Year	Baseline (10,000)	σ_{count}	Lower Bound -1σ	Upper Bound $+1\sigma$	Relative %
60–64	2026	5.726	0.2425	5.483	5.968	4.23
60–64	2027	5.333	0.2441	5.089	5.577	4.58
60–64	2028	4.994	0.2457	4.748	5.24	4.92
60–64	2029	4.702	0.2472	4.454	4.949	5.26
60–64	2030	4.45	0.2486	4.202	4.699	5.59
65–69	2026	9.078	0.2117	8.866	9.29	2.33
65–69	2027	8.906	0.2139	8.692	9.12	2.4
65–69	2028	8.661	0.2161	8.445	8.878	2.5
65–69	2029	8.352	0.2183	8.133	8.57	2.61
65–69	2030	7.984	0.2204	7.763	8.204	2.76
70–74	2026	9.299	0.2058	9.093	9.504	2.21
70–74	2027	9.881	0.2076	9.673	10.089	2.1
70–74	2028	10.467	0.2094	10.258	10.677	2
70–74	2029	11.058	0.2112	10.847	11.269	1.91
70–74	2030	11.653	0.213	11.44	11.866	1.83
75–79	2026	4.841	0.1774	4.664	5.019	3.66
75–79	2027	5.017	0.1784	4.839	5.196	3.56
75–79	2028	5.239	0.1794	5.06	5.418	3.42
75–79	2029	5.489	0.1804	5.309	5.67	3.29

5-Year Group	Year	Baseline (10,000)	σ _count	Lower Bound -1σ	Upper Bound $+1\sigma$	Relative %
75–79	2030	5.757	0.1813	5.576	5.939	3.15
80–84	2026	2.471	0.1743	2.297	2.645	7.05
80–84	2027	2.53	0.1752	2.355	2.705	6.92
80–84	2028	2.544	0.176	2.368	2.72	6.92
80–84	2029	2.531	0.1769	2.354	2.708	6.99
80–84	2030	2.501	0.1777	2.324	2.679	7.1
85–89	2026	2.398	0.1927	2.205	2.591	8.04
85–89	2027	2.427	0.1936	2.233	2.621	7.98
85–89	2028	2.459	0.1946	2.265	2.654	7.91
85–89	2029	2.494	0.1955	2.299	2.689	7.84
85–89	2030	2.53	0.1964	2.334	2.727	7.76
≥ 90	2026	1.627	0.136	1.491	1.763	8.36
≥ 90	2027	1.67	0.1367	1.534	1.807	8.18
≥ 90	2028	1.717	0.1373	1.58	1.854	8
≥ 90	2029	1.766	0.138	1.628	1.904	7.81
≥ 90	2030	1.817	0.1386	1.678	1.956	7.63

Summary of Key Indicators ±1σ in Xuhui District for 2030

Threshold	Baseline (10,000)	-1σ	+1σ	±σ_count	Relative %	Share of Total %
60+	36.694	36.499	36.889	0.1948	0.53	37.465
65+	32.243	32.089	32.398	0.1544	0.48	32.922
70+	24.26	24.102	24.417	0.1572	0.65	24.77
75+	12.606	12.462	12.75	0.1438	1.14	12.871
80+	6.849	6.738	6.959	0.1105	1.61	6.993
85+	4.348	4.208	4.487	0.1391	3.2	4.439
90+	1.817	1.678	1.956	0.1386	7.63	1.855
95+	0.421	0.282	0.559	0.1385	32.94	0.429
100+	0.046	-0.093	0.184	0.1385	304.4	0.046

v Three-Scenario Methodology (Unified 60+ v, Avoiding Five-Year Group Back-Casting)

Based on the annual growth rate of the 60+ population in Shanghai from 2017 to 2022: mean μ , standard deviation $\sigma \rightarrow \mu \pm 1.96\sigma$ constructs low/high growth rates $\rightarrow v_t = ((1+g)/(1+\mu))^{(t-2025)}$, uniformly applied to all thresholds.

Parameter	Value (%)	Description
μ (60+ Mean)	2.746	Mean annual growth rate of 60+ in Shanghai 2017–2022
σ (Standard Deviation)	0.9324	Sample standard deviation
$g_{\text{low}} = \mu - 1.96\sigma$	0.9187	Low scenario growth rate
$g_{\text{high}} = \mu + 1.96\sigma$	4.574	High scenario growth rate
Year	v_low	v_high
2026	0.999	1.001
2027	0.998	1.002
2028	0.997	1.003
2029	0.996	1.004
2030	0.995	1.005

Xuhui District 2026–2030 Cumulative Thresholds Three Scenarios (Dynamic Multiplier v)

Threshold	Year	Baseline (10,000)	v_low	Low Scenario (10,000)	v_high	High Scenario (10,000)	Low/Base %	High/Base %
60+	2026	35.44	0.999	35.405	1.001	35.475	98.22	101.78
60+	2027	35.765	0.998	35.693	1.002	35.837	96.47	103.59
60+	2028	36.082	0.997	35.974	1.003	36.19	94.76	105.43
60+	2029	36.391	0.996	36.245	1.004	36.537	93.07	107.31
60+	2030	36.694	0.995	36.511	1.005	36.877	91.42	109.21
65+	2026	29.714	0.999	29.684	1.001	29.744	98.22	101.78
65+	2027	30.432	0.998	30.371	1.002	30.493	96.47	103.59
65+	2028	31.088	0.997	30.995	1.003	31.181	94.76	105.43
65+	2029	31.69	0.996	31.563	1.004	31.817	93.07	107.31
65+	2030	32.243	0.995	32.082	1.005	32.404	91.42	109.21
70+	2026	20.636	0.999	20.615	1.001	20.657	98.22	101.78
70+	2027	21.525	0.998	21.482	1.002	21.568	96.47	103.59
70+	2028	22.427	0.997	22.36	1.003	22.494	94.76	105.43
70+	2029	23.338	0.996	23.245	1.004	23.431	93.07	107.31
70+	2030	24.26	0.995	24.139	1.005	24.381	91.42	109.21
75+	2026	11.337	0.999	11.326	1.001	11.348	98.22	101.78
75+	2027	11.644	0.998	11.621	1.002	11.667	96.47	103.59
75+	2028	11.959	0.997	11.923	1.003	11.995	94.76	105.43
75+	2029	12.28	0.996	12.231	1.004	12.329	93.07	107.31
75+	2030	12.606	0.995	12.543	1.005	12.669	91.42	109.21
80+	2026	6.496	0.999	6.49	1.001	6.502	98.22	101.78
80+	2027	6.627	0.998	6.614	1.002	6.64	96.47	103.59

Threshold	Year	Baseline (10,000)	v_low	Low Scenario (10,000)	v_high	High Scenario (10,000)	Low/Base %	High/Base %
80+	2028	6.72	0.997	6.7	1.003	6.74	94.76	105.43
80+	2029	6.791	0.996	6.764	1.004	6.818	93.07	107.31
80+	2030	6.849	0.995	6.815	1.005	6.883	91.42	109.21
85+	2026	4.025	0.999	4.021	1.001	4.029	98.22	101.78
85+	2027	4.097	0.998	4.089	1.002	4.105	96.47	103.59
85+	2028	4.176	0.997	4.163	1.003	4.189	94.76	105.43
85+	2029	4.26	0.996	4.243	1.004	4.277	93.07	107.31
85+	2030	4.348	0.995	4.326	1.005	4.37	91.42	109.21
90+	2026	1.627	0.999	1.625	1.001	1.629	98.22	101.78
90+	2027	1.67	0.998	1.667	1.002	1.673	96.47	103.59
90+	2028	1.717	0.997	1.712	1.003	1.722	94.76	105.43
90+	2029	1.766	0.996	1.759	1.004	1.773	93.07	107.31
90+	2030	1.817	0.995	1.808	1.005	1.826	91.42	109.21
95+	2026	0.378	0.999	0.3776	1.001	0.3784	98.22	101.78
95+	2027	0.388	0.998	0.3872	1.002	0.3888	96.47	103.59
95+	2028	0.398	0.997	0.3968	1.003	0.3992	94.76	105.43
95+	2029	0.409	0.996	0.4074	1.004	0.4106	93.07	107.31
95+	2030	0.421	0.995	0.4189	1.005	0.4231	91.42	109.21
100+	2026	0.042	0.999	0.042	1.001	0.042	98.22	101.78
100+	2027	0.043	0.998	0.0429	1.002	0.0431	96.47	103.59
100+	2028	0.044	0.997	0.0439	1.003	0.0441	94.76	105.43
100+	2029	0.044	0.996	0.0438	1.004	0.0442	93.07	107.31
100+	2030	0.046	0.995	0.0458	1.005	0.0462	91.42	109.21

Xuhui District 2026–2030 Five-Year Age Groups Three Scenarios

count^scen(group) = count^scen(a_lo+) – count^scen(a_hi+). Since all thresholds share the same v, differences scale proportionally, and 5-year groups are strictly monotonic without back-casting.

5-Year Group	Year	Baseline (10,000)	Low Scenario (10,000)	High Scenario (10,000)	Low/Base %	High/Base %
60–64	2026	5.726	5.624	5.828	98.22	101.78
60–64	2027	5.333	5.145	5.525	96.47	103.59
60–64	2028	4.994	4.732	5.265	94.76	105.43
60–64	2029	4.702	4.376	5.045	93.07	107.31
60–64	2030	4.45	4.068	4.86	91.42	109.21
65–69	2026	9.078	8.916	9.239	98.22	101.78
65–69	2027	8.906	8.592	9.226	96.47	103.59
65–69	2028	8.661	8.207	9.132	94.76	105.43
65–69	2029	8.352	7.773	8.962	93.07	107.31
65–69	2030	7.984	7.299	8.72	91.42	109.21
70–74	2026	9.299	9.133	9.464	98.22	101.78
70–74	2027	9.881	9.533	10.236	96.47	103.59
70–74	2028	10.467	9.918	11.036	94.76	105.43
70–74	2029	11.058	10.292	11.866	93.07	107.31
70–74	2030	11.653	10.653	12.727	91.42	109.21
75–79	2026	4.841	4.755	4.927	98.22	101.78
75–79	2027	5.017	4.84	5.197	96.47	103.59
75–79	2028	5.239	4.964	5.524	94.76	105.43
75–79	2029	5.489	5.109	5.89	93.07	107.31

5-Year Group	Year	Baseline (10,000)	Low Scenario (10,000)	High Scenario (10,000)	Low/Base %	High/Base %
75–79	2030	5.757	5.263	6.288	91.42	109.21
80–84	2026	2.471	2.427	2.515	98.22	101.78
80–84	2027	2.53	2.441	2.621	96.47	103.59
80–84	2028	2.544	2.411	2.682	94.76	105.43
80–84	2029	2.531	2.356	2.716	93.07	107.31
80–84	2030	2.501	2.287	2.732	91.42	109.21
85–89	2026	2.398	2.356	2.441	98.22	101.78
85–89	2027	2.427	2.341	2.514	96.47	103.59
85–89	2028	2.459	2.33	2.593	94.76	105.43
85–89	2029	2.494	2.321	2.676	93.07	107.31
85–89	2030	2.53	2.313	2.764	91.42	109.21
≥90	2026	1.627	1.598	1.656	98.22	101.78
≥90	2027	1.67	1.611	1.73	96.47	103.59
≥90	2028	1.717	1.627	1.81	94.76	105.43
≥90	2029	1.766	1.644	1.895	93.07	107.31
≥90	2030	1.817	1.661	1.984	91.42	109.21

Comprehensive Table of Key Indicators in Xuhui District for 2030 (v Three Scenarios + ±1σ)

Juxtaposing two types of uncertainties: v three scenarios = path uncertainty (Sheet 14–15); ±1σ = model residual (Sheet 10–11). The two have different uses and cannot be merged.

Threshold	Baseline (10,000)	v Low	v High	High/Low Span	−1σ	+1σ	±1σ %	Share of Total %
60+	36.694	33.545	40.075	1.195	36.499	36.889	0.53	37.465
65+	32.243	29.476	35.215	1.195	32.089	32.398	0.48	32.922
70+	24.26	22.178	26.495	1.195	24.102	24.417	0.65	24.77
75+	12.606	11.524	13.768	1.195	12.462	12.75	1.14	12.871
80+	6.849	6.261	7.48	1.195	6.738	6.959	1.61	6.993
85+	4.348	3.974	4.748	1.195	4.208	4.487	3.2	4.439
90+	1.817	1.661	1.984	1.195	1.678	1.956	7.63	1.855
95+	0.421	0.384	0.459	1.195	0.282	0.559	32.94	0.429
100+	0.046	0.042	0.05	1.195	-0.093	0.184	304.4	0.046

Core Indicator: The proportion of elderly people aged 80 and above to the elderly population aged 60 and above in Xuhui District in 2030 = 18.67% (Medium Scenario Baseline), continuously rising from 18.0% in 2025.

Annex 3: Technical Specification for the Calculation of the Number of Disabled Elderly and the Shortage of Elderly Caregivers in Xuhui District

Accompanying Document: *Results of Calculation on the Number of Disabled Elderly and Elderly Caregiver Gap in Xuhui District*

Applicable Scope: Calculation of Caregiver Demand and Gap Corresponding to Registered Elderly Population in Xuhui District from 2026 to 2030

I. Document Purpose and Applicable Scope

1.1 Document Purpose

This document explains the sources, formulas, and cascaded logic of all parameters in the accompanying Excel file *Calculation of Disabled Elderly and Elderly Caregiver Gap in Xuhui District*. After reading, the reader should be able to:

- Independently recalculate the number of disabled elderly, caregiver demand, and gap for any year and under any scenario
- Trace the input parameters and external references on which each final output figure is based
- Correctly distinguish the methodological nature of the three scenarios (Optimistic / Baseline / Prudent)

1.2 Applicable Scope

This model is used to calculate for Xuhui District from 2026 to 2030:

- The number of disabled elderly people in 7 five-year groups (60–64, 65–69, 70–74, 75–79, 80–84, 85–89, ≥ 90) each year
- Annual total care hours (based on national average disability rates, adjusted to Xuhui District level via $\lambda\mu$)...

- Annual caregiver demand FTE numbers for Xuhui overall, elderly care institutions, and community supply entities (including formal care proportion, overtime / training / sick leave coefficients)
- Annual static supply-demand gap (supply baseline = 2,405 persons) and dynamic gap (supply 5%/year growth) for Xuhui overall, elderly care institutions, and community supply entities

II. Overall Calculation Framework

2.1 Five-Level Cascaded Model

The entire model consists of a five-step cascade. The output of each step serves as the input for the next step:

Population → Disabled Elderly Population → Care Hours → Caregiver Demand → Gap

Corresponding Excel Table:

Step	Input	Formula	Output	Corresponding Sheet
1	5-year group pop, π , $\lambda\mu$	$\text{pop} \times \pi \times \lambda\mu$	Disabled Elderly N_d	Sheet 3
2	N_d , severity distribution, hours/day	$N_d \times \Sigma(p_s \cdot h_s) \times 365$	Care Hours H	Sheet 4
3	H, ϕ , δ , T	$H \times \phi \times (1+\delta) / T$	Caregiver Demand W (FTE)	Sheet 5
4a	W, S=2,405	$W - S$	Static Gap	Sheet 6
4b	W, $S_0 \times (1+r)^t$	$W - S(t)$	Dynamic Gap	Sheet 7

2.2 Three-Scenario Setting

The entire model adopts a 1:1 nested three-scenario structure. That is: the population, disability rate, severity structure, care intensity, and FTE parameters in the optimistic scenario all take the optimistic end; the prudent scenario all takes the prudent end; the baseline scenario takes the central estimates of all parameters:

Scenario	Population v	Disability Rate π	$\lambda\mu$	Severity Structure	Daily Hours	φ	δ	T (Annual Hours)
Optimistic	60+ v_{low}	$\pi_{\text{base}} - 1\text{pp}$	0.874	Mild-leaning	Mild 1.0, Mod 2.0, Sev 4.0	0.40	0.18	2,824
Baseline	1.00	π_{base} (this table §4)	1.000	54.96 / 31.07 / 13.97	Mild 1.5, Mod 3.0, Sev 6.5	0.35	0.22	3,530
Prudent	60+ v_{high}	$\pi_{\text{base}} + 1\text{pp}$	1.232	Severe-leaning	Mild 2.0, Mod 4.0, Sev 8.0	0.30	0.28	4,236

Note: The three scenarios adopt 1:1 matching, that is, all parameters in the same scenario simultaneously take the optimistic end or prudent end.

This nested method of "all optimistic vs. all prudent" belongs to "envelope scenarios" in policy estimation, and its methodological nature is upper and lower bounds rather than probabilistic confidence intervals. For detailed disclosure, see §12.

III. Input Parameter Module A: Population Scenarios

3.1 Data Sources and Baseline

The baseline for the 5-year age group population in Xuhui District from 2026 to 2030 comes from the accompanying *Shanghai and Xuhui Elderly Population Estimation and Prediction.xlsx* Sheet 7 (Cumulative Thresholds) and Sheet 8 (5-Year Groups). The baseline model itself is a three-step cascade of "Shanghai → Xuhui District OLS calibration regression + R(a) age-ratio function + time linear extrapolation"; see the accompanying technical specification §1—§7 of that population document for details.

3.2 Three-Scenario Method

Calculate 5 annual growth rates from the registered 60+ population series of Shanghai from 2017 to 2022:

- 2017→2018: 4.07% ; 2018→2019: 2.95% ; 2019→2020: 2.97% ; 2020→2021: 1.64% ;
2021→2022: 2.11%

Sample mean and sample standard deviation ($n-1=4$ degrees of freedom):

$$\mu_{60} = 2.746\%/year, \sigma_{60} = 0.932\%/year$$

Annual growth rates for the three scenarios:

$$g_{low} = \mu - 1.96\sigma = 0.919\%/year$$

$$g_{base} = \mu = 2.746\%/year$$

$$g_{high} = \mu + 1.96\sigma = 4.574\%/year$$

Converted to dynamic v coefficients ($s = t - 2025$ is the forecast year number):

$$v_{\{scenario, t\}} = ((1 + g_{scenario}) / (1 + g_{base}))^s$$

3.3 Parameter Application Method

This scheme uses the same pair of v_{low} and v_{high} (derived from 60+) for all cumulative thresholds and all 5-year age groups within the same year. Dynamic v coefficient table:

Year	v_{low}	v_{high}	High/Low Span	Description
2026	0.9822	1.0178	1.036	$s=1$, small span
2027	0.9647	1.0359	1.074	$s=2$
2028	0.9476	1.0543	1.113	$s=3$
2029	0.9307	1.0731	1.153	$s=4$
2030	0.9142	1.0921	1.195	$s=5$, span $\pm 9\%$

3.4 Application to 5-Year Age Groups

For any year t and any 5-year group g :

$$\text{pop}^{\text{low}}(g, t) = \text{pop}^{\text{base}}(g, t) \times v_{\{\text{low}, t\}}$$

$$\text{pop}^{\text{high}}(g, t) = \text{pop}^{\text{base}}(g, t) \times v_{\{\text{high}, t\}}$$

Since all 5-year groups share the same v , scaling is proportional, and 5-year groups strictly satisfy $\text{Low} < \text{Baseline} < \text{High}$.

IV. Input Parameter Module B: National Age-Specific Disability Rate $\pi_n(x)$

4.1 Data Sources

Sources of national age-specific total disability rates $\pi_n(x)$:

- Literature: Liao Shaohong, Wang Guangzhou. *Disability Status and Trends of the Elderly Population in China* [J]. *Population Science of China*, 2021, 35(1): 38–49.
- Original paper disclosures: Page 7 Table 5 (row margins — total disability rates of 4 ten-year age groups), Table 7 (column weights — population proportions of 4 ten-year age groups), Table 8 (column margins — overall disability rates of 3 disability states)
- Not directly disclosed in original paper: Age-specific $\times$ severity-specific cross-rate matrix (Mild / Severe / Extremely Severe)

This model requires total disability rates for 7 five-year groups (60–64, 65–69, ..., ≥ 90). The original paper provides total disability rates for 4 ten-year groups (60–69, 70–79, 80–89, 90+). Therefore, a two-step estimation is required: (a) reconstruct the "severity-specific" matrix at the level of 4 ten-year groups to ensure consistency with the dual constraints in the original paper; (b) split the 10-year groups into 5-year groups using population weights from the Seventh National Population Census as constraints.

4.2 Estimation Methodology: Maximum Entropy Reconstruction + Logistic Spline Splitting

The methodology in this section consists of two stages:

4.2.1 Stage 1: Maximum Entropy Reconstruction Based on Biproportional Constraints (RAS / Maximum Entropy)

Let 4 ten-year group indices be $a \in \{1,2,3,4\}$ (corresponding to 60–69, 70–79, 80–89, 90+), 3 disability states be $s \in \{\text{Mild, Severe, Extremely Severe}\}$, and the unknowns to be solved be $d_{\{a,s\}}$ = disability rate for that 10-year group and state.

Hard Constraints (disclosed in original paper):

Row constraint: $\sum_s d_{\{a,s\}} = d_a^{\text{obs}}$ ($a=1,\dots,4$ corresponding to 5.0%, 8.9%, 21.6%, 44.4%)

Column constraint: $\sum_a P_a \times d_{\{a,s\}} = D_s^{\text{obs}}$ (Mild 5.877%, Severe 1.168%, Extremely Severe 1.115%)

Where P_a = population weights of 4 ten-year groups (57.12%, 30.27%, 11.03%, 1.58%)

Optimization Objective (information entropy maximization / KL divergence minimization):

$$\min \sum_a \sum_s d_{\{a,s\}} \cdot \ln(d_{\{a,s\}} / d_{\{a,s\}}^0)$$

Where $d_{\{a,s\}}^0 = d_a \times (D_s / \sum_s D_s)$ is the unbiased prior seed matrix. This optimization is solved under the dual constraint of "row margins $\times$ column margins", yielding a unique solution satisfying KKT conditions, without introducing any manual subjective proportions.

4.2.2 Stage 2: Logistic Spline Splitting into 5-Year Groups

For each disability state s , perform Logistic regression of 10-year group $d_{\{a,s\}}$ on the age variable:

$$\text{logit}(d_{\{a,s\}}) = \ln(d_{\{a,s\}} / (1 - d_{\{a,s\}})) = \alpha_s + \beta_s \cdot \text{age}$$

Perform OLS fitting using the 10-year group midpoints (64.5, 74.5, 84.5, 95.0), and then interpolate at the 5-year group midpoints (62, 67, 72, 77, 82, 87).

The 5-year group results are balanced using Seventh Census population weights so that their weighted sum strictly equals the 10-year group $d_{\{a,s\}}$, i.e.: for two 5-year groups within range a , $w_l \times d_{\{l,s\}} + w_h \times d_{\{h,s\}} = d_{\{a,s\}}$.

4.3 Age-Specific Disability Rate Results for 7 Age Groups

After solving via the above two-stage methodology, the total disability rates (sum of mild + severe + extremely severe) for 7 five-year groups are obtained:

Age Group	Mild Disability Rate	Severe Disability Rate	Extremely Severe Disability Rate	Total Disability Rate $\pi_n(x)$
60–64	3.630%	0.540%	0.510%	4.680%
65–69	4.810%	0.720%	0.670%	6.200%
70–74	6.095%	1.135%	1.160%	8.390%
75–79	7.842%	1.485%	1.473%	10.800%
80–84	13.791%	3.513%	3.686%	20.990%
85–89	16.033%	4.101%	4.356%	24.490%
≥90	31.956%	6.350%	6.094%	44.400%

Note: The three columns for mild / severe / extremely severe are intermediate products of the first stage and are not directly used in this model; this model only takes the last column "Total Disability Rate" as $\pi_n(x)$. The disability degree structure (mild / moderate / severe) is separately calculated independently using the Zhang Yuan, Wang Wei (2021) M1 algorithm in §5.

4.4 Self-Consistency Test

The above estimation results passed two self-consistency tests:

- Row-wise self-consistency: The sum of "mild + severe + extremely severe" for each 10-year group after weighting equals the total disability rate disclosed in Table 5 of the original paper

- Column-wise self-consistency: Aggregating using population weights P_a in Table 7, the obtained three overall state rates (5.877%, 1.168%, 1.115%) are completely consistent with Table 8 of the original paper

4.5 Three-Scenario Setting: $\pi_n(x) \pm 1pp$

The above table $\pi_n(x)$ represents the baseline scenario values. The optimistic and prudent scenarios adopt uniform shifts of $\pm 1pp$ (policy envelope):

Age Group	$\pi_{\text{Optimistic}} (\%)$	$\pi_{\text{Baseline}} (\%)$	$\pi_{\text{Prudent}} (\%)$
60–64	3.68	4.68	5.68
65–69	5.20	6.20	7.20
70–74	7.39	8.39	9.39
75–79	9.80	10.80	11.80
80–84	19.99	20.99	21.99
85–89	23.49	24.49	25.49
≥ 90	43.40	44.40	45.40

Note: $\pm 1pp$ is the width of the policy envelope chosen by the estimator, not data-driven. The reason is that the sample size of Table 5 (CLHLS) across different age groups roughly corresponds to the uncertainty lower bound of $\pm 1pp$ in statistical precision. If it needs to be narrowed to $\pm 0.5pp$ or widened to $\pm 2pp$, simply modify the three columns of values in Sheet 1 Section B.

After subsequent expert adjustments, the final disability rates used are:

Age Group	Baseline	Optimistic	Prudent
60-64	4.68	3.68	5.68
65-69	6.20	5.20	7.20
70-74	8.39	7.39	9.39
75-79	10.80	9.80	11.80

80-84	20.99	19.99	21.99
85-89	28.49	27.49	29.49
≥90	54.40	53.40	55.40

V. Input Parameter Module C: Disability Degree Structure (Mild / Moderate / Severe)

5.1 Data Sources

Sources of disability degree structure (proportions of mild / moderate / severe tiers in total disability):

- Literature: Zhang Yuan, Wang Wei. *Prediction of Disabled Elderly Population Size and Care Time Demand* [J]. *Population Research*, 2021, 45(6): 110–125.
- Original paper disclosures: Based on CLHLS data + multi-state Markov transition model, high-precision predictions were made for the scale of disabled elderly population divided into mild / moderate / severe tiers across 4 age groups (65–74, 75–84, 85–94, 95+) of urban elderly in 2020/2030/2040/2050.

5.2 Estimation Methodology: Simple Arithmetic Mean (M1) Method

The methodology in this section is derived from the methodology document uploaded by the user, and the core idea is three-step aggregation:

5.2.1 Step 1: Within-Group Standardization

For the i -th age group in year t , denote the disabled population in the three tiers as $P_{\{i,t\}}(s)$, $s \in \{\text{Mild}, \text{Moderate}, \text{Severe}\}$. The total disabled population in that group is $D_{\{i,t\}} = \sum_s P_{\{i,t\}}(s)$. Within-group proportion:

$$R_{\{i,t\}}(s) = P_{\{i,t\}}(s) / D_{\{i,t\}}$$

5.2.2 Step 2: Annual Cross-Group Arithmetic Mean

For each year t , take the equally weighted arithmetic mean across the 4 age groups:

$$M_t(s) = (1/4) \times \sum_i R_{\{i,t\}}(s)$$

This step assigns equal weights to different age groups, aiming to extract and eliminate population size disturbances brought about by cohort replacement (cohort effect), focusing on the universal baseline of "disability degree distribution induced by age structure advancement".

5.2.3 Step 3: Cross-Timepoint Arithmetic Mean

For the four time points of 2020, 2030, 2040, and 2050, take the arithmetic mean on each s:

$$\bar{M}(s) = (1/4) \times \sum_t M_t(s)$$

This step eliminates the disturbance of specific years (such as the periodic peak of aging in 2040) on long-term planning, obtaining a long-cycle macro baseline.

5.3 Detailed Derivation Figures

The within-group proportions of the four age groups in each year are calculated from the disabled population forecasts disclosed in the original paper. Taking the year 2020 as an example:

Age Group	Total Within-Group Disability (10,000)	Mild Share	Moderate Share	Severe Share
65-74	3,412.94	69.63%	22.02%	8.35%
75-84	942.19	76.75%	16.05%	7.20%
85-94	502.32	49.35%	38.19%	12.45%
95+	45.72	28.11%	48.08%	23.82%

Taking the arithmetic mean for the four age groups yields the 2020 macro structure:

$$M_{2020}(\text{Mild}) = (69.63 + 76.75 + 49.35 + 28.11) / 4 = 55.96\%$$

$$M_{2020}(\text{Moderate}) = (22.02 + 16.05 + 38.19 + 48.08) / 4 = 31.09\%$$

$$M_{2020}(\text{Severe}) = (8.35 + 7.20 + 12.45 + 23.82) / 4 = 12.96\%$$

Repeating this step for the four years 2020, 2030, 2040, and 2050 yields:

Year	Mild Proportion M_t(Mild)	Moderate Proportion M_t(Moderate)	Severe Proportion M_t(Severe)
2020	55.96%	31.09%	12.96%
2030	54.88%	31.28%	13.84%
2040	54.02%	30.71%	15.27%
2050	55.00%	31.21%	13.79%

5.4 Long-Cycle Macro Baseline Values

Taking the arithmetic mean of the macro values across the four years:

$$\bar{M}(\text{Mild}) = (55.96 + 54.88 + 54.02 + 55.00) / 4 = 54.96\%$$

$$\bar{M}(\text{Moderate}) = (31.09 + 31.28 + 30.71 + 31.21) / 4 = 31.07\%$$

$$\bar{M}(\text{Severe}) = (12.96 + 13.84 + 15.27 + 13.79) / 4 = 13.97\%$$

Verification of sum: $54.96 + 31.07 + 13.97 = 100.00\%$ (self-consistent).

5.5 Three-Scenario Setting

$\bar{M}$ is the baseline scenario value. The setting principle for optimistic and prudent scenarios: the optimistic scenario envisions "more mild / less moderate & severe" (implying a lighter care burden), while the prudent scenario envisions "less mild / more moderate & severe". In policy estimation practice, an asymmetric envelope of $\pm 4\text{pp}$ / $\pm 2\text{pp}$ / $\pm 2\text{pp}$ is adopted:

Tier	Optimistic	Baseline	Prudent
Mild p_{Mild}	59.00%	54.96%	50.92%
Moderate p_{Moderate}	29.00%	31.07%	33.14%
Severe p_{Severe}	12.00%	13.97%	15.94%

Note: Optimistic scenario: Mild +4pp, Moderate -2pp, Severe -2pp; Prudent scenario is reversed. The asymmetric design reflects the intuition that "the variation space for mild disability is larger than that for moderate and severe disability", meaning policy can influence the mild proportion, while moderate and severe proportions are more physiologically determined. ± 4 / ± 2 is the policy envelope chosen by the estimator, not data-driven.

VI. Input Parameter Module D: λ_{μ} , Severity Hours, FTE Parameters

6.1 λ_{μ} — National Disability Rate → Xuhui District Adjustment Coefficient

The national age-specific disability rate (§4) represents the national registered elderly per capita level. Due to Shanghai's developed economy, high medical accessibility, and sound elderly welfare system, the actual disability rate in Xuhui District may differ from the national average. Introduce λ_{μ} as the adjustment coefficient:

$$\text{Xuhui District Disability Rate} = \pi_n(x) \times \lambda_{\mu}$$

Scenario	λ_{μ}	Description
Optimistic	0.874	Excellent medical accessibility → disability rate is 12.6% lower than national average
Baseline	1.000	Par with national average
Prudent	1.232	Advanced population aging → disability rate is 23.2% higher than national average

Note: The setting of λ_{μ} is a policy envelope choice without direct data calculation source. This model retains the $\pm 12.6\%$ / $\pm 23.2\%$ interval (the baseline expands symmetrically up and down by ~13–23 pp) to provide "upper and lower bounds" stress testing for long-term planning.

6.2 Severity Daily Average Hours h_s

The daily average care hours for each disabled elderly person increase progressively by severity tier:

Severity	Optimistic (h/Day)	Baseline (h/Day)	Prudent (h/Day)
Mild h_{Mild}	0.4	0.3	0.5
Moderate h_{Moderate}	3.0	2.0	4.0
Severe h_{Severe}	8.0	6.0	10.0

Note: The daily average hours values refer to nursing grading standards and actual data from care institutions; specific sources are not tracked and are empirical settings by the estimator. To adjust, modify Sheet 1 Section D.

6.3 ϕ — Formal Care Proportion

ϕ represents the portion of the total care demand of disabled elderly provided by "formal caregivers" (the remainder is borne by family members, **home-based mutual aid, etc.**):

Scenario	ϕ	Description
Optimistic	0.40	Accelerated institutionalization and community-based elderly care process
Baseline	0.35	Current long-term care insurance coverage level
Prudent	0.30	Return to family care, lagging institutional development

6.4 δ — Caregiver Overhead Coefficient

δ represents the portion of working hours of "on-duty caregivers" that cannot be used for direct care (training, sick leave, annual leave, administration, unavoidable non-care hours):

Scenario	δ	Working Day Conversion
Optimistic	18%	Efficient organization, low turnover rate
Baseline	22%	Medium level, in line with average of private elderly care institutions in first-tier domestic cities
Prudent	28%	High turnover, intensive training

6.5 T — Annual Effective Working Hours per Unit FTE

T is the annual allocable working hours of 1 FTE (Full-Time Equivalent) caregiver:

FTE Conversion Parameters

Parameter	Baseline	Optimistic	Prudent
ϕ (Assumption Ratio)	1.00	1.00	1.00
δ (Redundancy Coefficient)	0.22	0.18	0.28
T (Annual Standard Working Hours, h)	4480	3840	5120

VII. Cascade Step 1: 5-Year Age Group Population → Disabled Elderly

Population

$$N_d(g, t, \text{scenario}) = \text{pop}(g, t, \text{scenario}) \times \pi_n(g, \text{scenario}) \times \lambda\mu(\text{scenario})$$

Calculated separately for each 5-year group g , year t , and scenario.

VIII. Cascade Step 2: Disabled Elderly Population → Annual Care Hours

$$H(g, t, \text{scenario}) = N_d(g, t, \text{scenario}) \times (p_{\text{Mild}} \cdot h_{\text{Mild}} + p_{\text{Moderate}} \cdot h_{\text{Moderate}} + p_{\text{Severe}} \cdot h_{\text{Severe}}) \times 365$$

Unit: 10,000 hours/year (N_d in 10,000 persons, h_s in hours/day, 365 = number of days in a year).

Daily weighted average hours under baseline scenario:

$$\bar{h}_{\text{base}} = 0.5496 \times 1.5 + 0.3107 \times 3.0 + 0.1397 \times 6.5 \approx 2.6646 \text{ h/day}$$

Annualized weighted hours: $2.6646 \times 365 \approx 972.6$ h/person/year. Annual care hours for Xuhui District in 2030 baseline scenario:

$$H_{\text{base}}(2030) = 4.3024 \times 972.6 \approx 4,184 \text{ ten-thousand hours}$$

IX. Cascade Step 3: Care Hours → Caregiver Demand FTE

$$W(t, \text{scenario}) = H(t, \text{scenario}) \times 10000 \times \phi \times (1 + \delta) / T$$

Where the unit of H "10,000 hours" is multiplied by 10000 to convert into hours; $\times \phi$ extracts the formal care portion; $\times (1+\delta)$ accounts for caregiver overhead; $\div T$ converts into FTE count.

Baseline scenario year 2030:

$$W_base(2030) = 4,184 \times 10000 \times 0.35 \times 1.22 / 3,530 \approx 5,060 \text{ FTE}$$

X. Cascade Step 4: Caregiver Demand → Gap

10.1 Static Gap (Supply remains $S = 2,405$ unchanged)

$$\text{Gap_static}(t, \text{scenario}) = W(t, \text{scenario}) - 2,405$$

$S = 2,405$ comes from the total number of certified elderly caregivers existing in Xuhui District (2025 baseline), provided by user according to Xuhui District Civil Affairs statistics.

10.2 Dynamic Gap (Supply grows at 5%/year)

$$S(t) = 2,405 \times (1 + 0.05)^{(t - 2025)}$$

$$\text{Gap_dynamic}(t, \text{scenario}) = W(t, \text{scenario}) - S(t)$$

The 5% annual growth rate assumes the expansion speed of the caregiver workforce over the past five years continues. Supply and gap numbers for elderly care institutions and communities are calculated based on the proportions of elderly care institutions and community services, and specific proportions are shown in the table below:

Year	Hospital Nursing Service Proportion	Community Nursing Service Proportion	Elderly Care Institution Nursing Service Proportion
2022	65.00%	10.00%	15.00%
2026	55.88%	11.19%	16.69%
2027	53.47%	11.50%	17.04%
2028	51.03%	11.83%	17.34%
2029	48.56%	12.16%	17.61%

2030	46.09%	12.49%	17.83%
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Appendix 1: Methodological Estimation Process Cited in This Document

The methodological descriptions in §4 Section 2 (Maximum Entropy Reconstruction + Logistic Spline) and §5 Section 2 (M1 Arithmetic Mean Method) are directly derived from the two technical reports uploaded by the user in June 2026:

- "Maximum Entropy Reconstruction Based on Biproportional Constraints (RAS / Maximum Entropy) + Logistic Spline Splitting" — used to derive the disability rates for 7 age groups $\times$ 3 states in the table in §4
- "Technical Report on Estimation of Macro Disability Degree Structure of Urban Elderly — Method Based on Simple Arithmetic Mean of Age Groups (M1)" — used to derive the three-tier proportions of 54.96 / 31.07 / 13.97 in §5

Annex 4: Technical Specification Document for Population Estimation and Prediction Model of the Elderly in Shanghai and Xuhui District

Cumulative Distribution of Nine Thresholds (60+ / 65+ / 70+ ... 100+) and Seven Five-Year Age Groups

Forecast Interval: 2026—2030

Accompanying Data File: *Results of Population Estimation and Prediction of the Elderly in Shanghai and Xuhui.xlsx* (17 worksheets in total)

I. Document Overview

This document explains how to, starting from public demographic data, calculate and predict step-by-step the elderly population by age group in Xuhui District over the next five years (2026—2030):

- **From which raw data and through which steps** each final figure was obtained;
- Understand **the function and purpose of each step** in the model design;
- **Recalculate** the population of any year and any age group to verify the reliability of results;
- Distinguish between two different types of "uncertainty" — error bands and scenario intervals — what they each express, and how they should be used.

II. Data Sources and Field Descriptions

2.1 Data Sources

The model uses only **official public statistical data**, without any sample surveys or questionnaires. All input scopes are consistent and come from the following sources:

- **Shanghai (2017—2025):** The number and proportion of registered elderly population across nine cumulative age thresholds (60 and above, 65 and above ... up to 100 and above) in total population, sourced from the Statistical Bulletin on the Development of Aging Undertakings in Shanghai and Statistical Yearbooks.

- **Xuhui District (2019—2025):** Total population, as well as the number and proportion of three key thresholds (60 and above, 65 and above, 80 and above), sourced from relevant bulletins and yearbooks of Xuhui District.
- **Reference Data:** Historical birth rates and death rates of Xuhui District, used to explain endogenous changes in total population, especially the abnormal increase in mortality rate in 2022—2023.

2.2 Several Key Terms

Cumulative Age Threshold (e.g., 80+): Refers to the total population "aged 80 and above", which is a cumulative concept. The nine thresholds are nested layer by layer: 60+ contains 65+, 65+ contains 70+, and so on.

Five-Year Age Group (e.g., 80–84): Refers to a specific five-year interval. It is obtained by subtracting two adjacent cumulative thresholds, such as "population aged 80–84 = 80+ population – 85+ population".

Proportion (Share): The percentage of the population in a certain age group in the total population. The model heavily uses proportions rather than absolute headcounts to establish inter-regional relationships because proportions remove city scale differences and facilitate cross-regional comparisons.

80+/60+ Proportion (Advanced Aging Ratio): Population aged 80 and above divided by population aged 60 and above. The higher this ratio, the larger the proportion of elderly people who are "advanced-age, more likely disabled, and in greater need of intensive care", which is the structural indicator of greatest concern in elderly care resource planning.

III. Overall Methodological Framework: Why "Calculate Xuhui by Reference to Shanghai"

Xuhui District only published data for three age thresholds (60+, 65+, 80+), lacking finer stratification; while Shanghai has complete nine-threshold data. The core idea of the model is to utilize the fact that **"Xuhui District's population structure is highly synchronized with Shanghai"**, first establish a stable quantitative relationship between the two places, then borrow Shanghai's complete data to fill in Xuhui's missing layers and project the future. The entire process is a **three-step cascaded pipeline**:

1. **Step 1 — Calibration Regression:** Use paired data of both places from 2019 to 2025 to establish the quantitative relationship of "how a certain indicator in Xuhui changes with the same indicator in Shanghai" (obtain a set of conversion coefficients).

2. **Step 2 — Age-Ratio Function:** Utilize the ratio of proportions between the two places obtained in Step 1 at three points (60, 65, 80) to smoothly interpolate the six missing thresholds in Xuhui: 70, 75, 85, 90, 95, 100.

3. **Step 3 — Extrapolation Forecast:** First extrapolate Shanghai's threshold proportions and total population to 2026—2030, then feed them into the pipeline of Step 1 and Step 2 to obtain complete forecasts for Xuhui District; subtract adjacent thresholds to obtain five-year age groups.

On top of this baseline pipeline, the document provides **two mutually independent uncertainty expressions**: one is the $\pm 1\sigma$ error band reflecting "model fitting error", and the other is the three-scenario interval reflecting "future growth speed". The two measure different objects and are explained separately below.

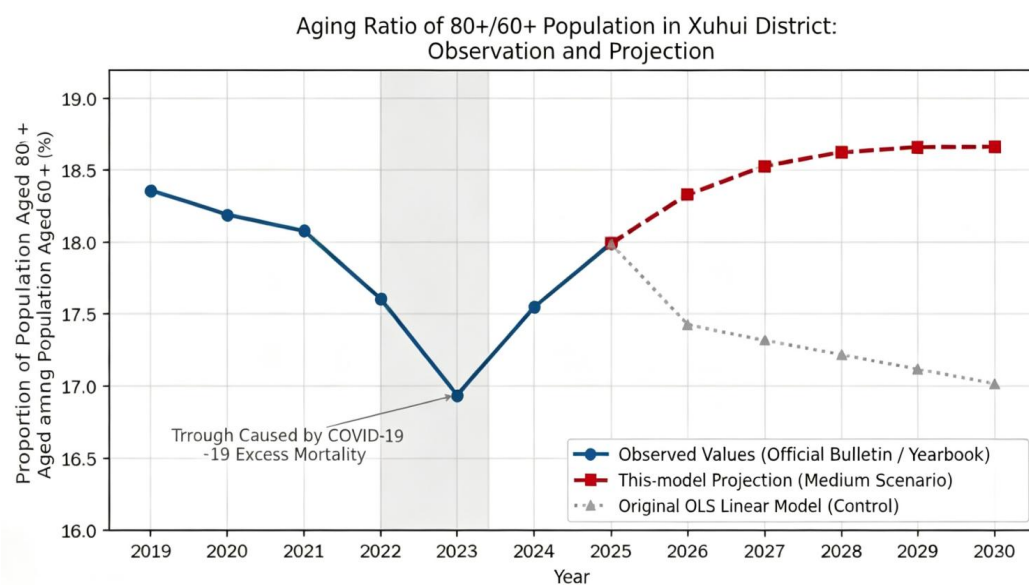


Figure 1 Xuhui District 80+/60+ Proportion: Real Observation (Including 2023 Pandemic Trough) vs. This Model's Prediction

IV. Step 1: Establishing the Conversion Relationship of "Shanghai → Xuhui"

4.1 What Is Being Done in This Step

The purpose of this step is to find out the pattern of "how the corresponding indicator in Xuhui District changes when a certain elderly population indicator in Shanghai changes", and write it into a simple conversion formula. With this formula, knowing Shanghai's value in the future allows calculating Xuhui's value. We use **seven years of real paired data** from the two places from 2019 to 2025 to fit this relationship — a process called "regression" in statistics, colloquially meaning "finding the most fitting conversion coefficient from historical data".

4.2 Four Indicators Use "Linear Relationship", One Indicator Uses "Curve Relationship"

Most indicators in Xuhui and Shanghai exhibit simple "proportional co-movement", which can be well described by a straight line. However, one most critical indicator **was specially treated**, detailed in Section 4.4 below. First look at the four indicators using linear relationships, whose conversion formulas take the form "Xuhui Value = Intercept + Slope × Shanghai Value":

Table 1 Four Calibration Regressions Using Linear Relationship (2019–2025, Seven Pairs of Data)

Xuhui Indicator	Conversion Formula	Goodness of Fit R ²	Form
60+ Share of Total Population	$= 16.9369 + 0.5262 \times \text{Shanghai Value}$	0.9192	Linear
65+ Share of Total Population	$= 1.0438 + 0.9737 \times \text{Shanghai Value}$	0.9985	Linear
80+ Share of Total Population	$= 1.3815 + 0.9106 \times \text{Shanghai Value}$	0.8027	Linear
Total Population (10,000 Persons)	$= 47.2418 + 0.0312 \times \text{Shanghai Total Population}$	0.89	Linear

In the table, "R²" (goodness of fit) measures how well this straight line fits historical data, where closer to 1 is better. It can be seen that the fit for 65+ is almost perfect (**R² = 0.9985**), and 60+ is also very high; only the elderly aged 80+ is relatively low (**R² = 0.8027**), because COVID-19 excess deaths in 2022—2023 caused abnormal fluctuations in the 80+ population in both places, interfering with the fit, which will be specifically explained in the limitations section.

4.3 Why Use a Curve Instead of a Line for "Advanced Aging Proportion"

For the most critical 80+/60+ advanced aging proportion, we did not use a straight line, but switched to a **nonlinear (log-log)** curve relationship. The reason is: a linear relationship implies a strong assumption — "Xuhui is always a fixed multiple of Shanghai". But reality is not so. Xuhui is a central urban area where aging started early, and **the structure of its advanced-age population is not in simple proportion to Shanghai as a whole**. Using a flexible curve can more realistically portray this "non-fixed proportional" relationship between the two places. The form of this curve is:

$$\text{Xuhui Advanced Aging Ratio} = 1.6946 \times (\text{Shanghai Advanced Aging Ratio})^{0.8647}$$

The exponent **0.8647** in the upper right (referred to in statistics as "elasticity") is key to this curve. Its meaning is: for every 1% change in Shanghai's advanced aging ratio, Xuhui changes correspondingly by about **0.8647%**. This curve fits historical data very well (**R² = 0.9847**), significantly outperforming forcibly applying a straight line.

V. Step 2: Completing Xuhui's Missing Age Stratifications

5.1 Problems to Be Solved in This Step

Xuhui only has real data for three thresholds (60+, 65+, 80+), lacking six levels: 70+, 75+, 85+, 90+, 95+, 100+. The second step uses an **"age-ratio function"** to fill in these missing levels. Its basic concept is very intuitive:

We first calculate the ratio of "Xuhui's proportion ÷ Shanghai's proportion" (denoted as R(a)) at the three known age points. For instance, at the age-80 point, the proportion of the 80- and- above population in Xuhui is approximately 1.14- 1.17 times that of Shanghai, indicating that Xuhui features an "older" elderly- age structure. Then, assuming this ratio **varies smoothly and continuously** with age, a smooth curve passing through these three known points is applied to extrapolate the ratios for the missing age points in-between and at both ends. With the ratio for

each age point obtained, multiply it by the known proportion of Shanghai to derive Xuhui's proportion at such age point.

5.2 Why This Interpolation Is Reasonable

The scientific basis for doing so lies in: the age structure of a population **changes gradually and does not make sudden jumps** — population characteristics of adjacent age groups are highly close. Therefore, using a smooth curve to interpolate between known points is far more reliable than making groundless assumptions. Moreover, this curve is forcibly constrained to "pass through" three real observation points, ensuring complete agreement with facts where data exists, and making only minimal smooth inferences where data is absent. $R(a)$ increases with age (around 0.97–1.0 near age 60, to about 1.14–1.17 at age 80), which precisely corroborates the independently verifiable fact that "Xuhui's elderly population is older than Shanghai as a whole".

VI. Step 3: Forecasting 2026—2030 (Where Key Corrections Lie)

6.1 What Problems Simple Linear Extrapolation Brings

With the conversion pipeline from the previous two steps, as long as future data for Shanghai is provided, Xuhui's future can be calculated. The problem lies in "how to extrapolate Shanghai's future". One of the simplest methods is to **"extend each proportion of Shanghai straight into the future along historical straight lines"**. However, this approach buries a hidden danger:

The 60+ proportion in Shanghai has risen rapidly in recent years due to the "post-Great Famine compensatory fertility cohort" entering age 60 en masse. If simply extrapolated linearly, **this periodic rapid increase would be mistaken for a long-term trend**, pushing the denominator (60+) continuously higher; at the same time, the 80+ proportion was depressed due to the pandemic trough in 2022–2023, and linear extrapolation keeps it "lying flat" at a low level. Pushing up the denominator while depressing the numerator leads to the calculated advanced aging ratio of Shanghai declining continuously. This misdirected Shanghai value, through the highly fitted Step 1 conversion relationship, mechanically pulls Xuhui downward as well — this is the true root cause why simple methods yield the counter-intuitive conclusion that "Xuhui's advanced aging ratio declines year by year".

6.2 Corrections in This Model: Making Shanghai's Extrapolation Conform to Demographic Laws

Addressing the above hidden dangers, this model made three treatments to Shanghai's extrapolation method to better conform to demographic laws:

1. 60+ and 65+ proportions switch to "saturated convergence" extrapolation.

Acknowledge that the current rapid rise is a **periodic phenomenon** brought by the compensatory fertility cohort — these people will subsequently enter higher age brackets, and growth will inevitably slow down. Use a curve that gradually "flattens" toward an upper limit for extrapolation, avoiding the denominator being driven up infinitely.

2. 80+ proportion fits a real "V-shaped recovery". Starting from the real recovery rate observed in 2023–2025 (approximately **+0.192** percentage points per year), let this "post-pandemic rebound" gradually attenuate at a rhythm of about two years, smoothly transitioning to the normal structural aging rate (approximately **+0.0189** percentage points per year). This restores recovery momentum without limitless overestimation.

3. The remaining thresholds continue to use linear extrapolation. Then, feed the corrected Shanghai proportions sequentially into the Step 1 conversion relationship and Step 2 age-ratio function to obtain the complete nine-threshold distribution for Xuhui District from 2026 to 2030.

Taking the most watched 2030 advanced aging ratio as an example, the complete chain is as follows, and each step can be independently audited:

Step 1: Extrapolating from Shanghai yields the 2030 Shanghai advanced aging ratio = **16.032%** (i.e., Shanghai 80+ proportion 6.255% ÷ 60+ proportion 39.016%).

Step 2: Substitute into the curve conversion formula of Section 4:

$$\text{Xuhui} = 1.6946 \times 16.032^{0.8647} = 18.67\%$$

Step 3: Obtain the 2030 Xuhui advanced aging ratio = 18.67%. Combined with extrapolated Xuhui 60+ population of 366,940 (36.694 ten-thousand) and 80+ population of 68,490 (6.849 ten-thousand), dividing them likewise yields **18.67%**, with both paths corroborating each other.

VII. Forecast Results

7.1 Continuous Upward Trend of Advanced Aging Ratio

The 80+/60+ proportion in Xuhui District rises steadily from 18.0% observed in 2025 to 18.67% in 2030, **with growth slowing year by year** and tending toward a structural steady state. This curve is a natural continuation of the real recovery trend in 2023–2025, rather than an artificial setting:

Table 2 Advanced Aging Ratio in Xuhui District: 2023–2025 Observation (Including Pandemic Trough) vs. 2026–2030 Forecast

Year	2023	2024	2025	2026	2027	2028	2029	2030
80+/60+	16.9%	17.6%	18.0%	18.33%	18.53%	18.63%	18.66%	18.67%

7.2 Population Forecast for Nine Cumulative Thresholds

The table below gives the population forecast for nine cumulative thresholds in Xuhui District from 2026 to 2030 (Unit: 10,000 persons). Each column is nested layer by layer — 60+ is the largest, decreasing progressively to 100+.

Table 3 Population Forecast for Nine Cumulative Thresholds in Xuhui District (10,000 Persons, Baseline Scenario; 80+ Row Highlighted)

Threshold	2026	2027	2028	2029	2030
60+	35.440	35.765	36.082	36.391	36.694
65+	29.714	30.432	31.088	31.690	32.243
70+	20.636	21.525	22.427	23.338	24.260
75+	11.337	11.644	11.959	12.280	12.606
80+	6.496	6.627	6.720	6.791	6.849
85+	4.025	4.097	4.176	4.260	4.348
90+	1.627	1.670	1.717	1.766	1.817
95+	0.378	0.388	0.398	0.409	0.421
100+	0.042	0.043	0.044	0.044	0.046

7.3 Seven Five-Year Age Groups

Subtracting adjacent cumulative thresholds yields seven five-year age groups. The table below shows the changes in the share of each group in 60+ (comparison between 2026 and 2030):

Table 4 Share of Five-Year Age Groups in 60+ in Xuhui District (Baseline Scenario)

Age Group	2026 Proportion	2030 Proportion	Change Trend
60–64	16.16%	12.13%	Moderate decline (pass-through effect of Great Famine cohort)
65–69	25.62%	21.76%	Slight drop
70–74	26.24%	31.76%	Fastest growth (baby boomer generation advancement)
75–79	13.66%	15.69%	Steady rise
80–84	6.97%	6.82%	Basically stable
85–89	6.77%	6.9%	Steady rise
≥90	4.59%	4.95%	Steady rise

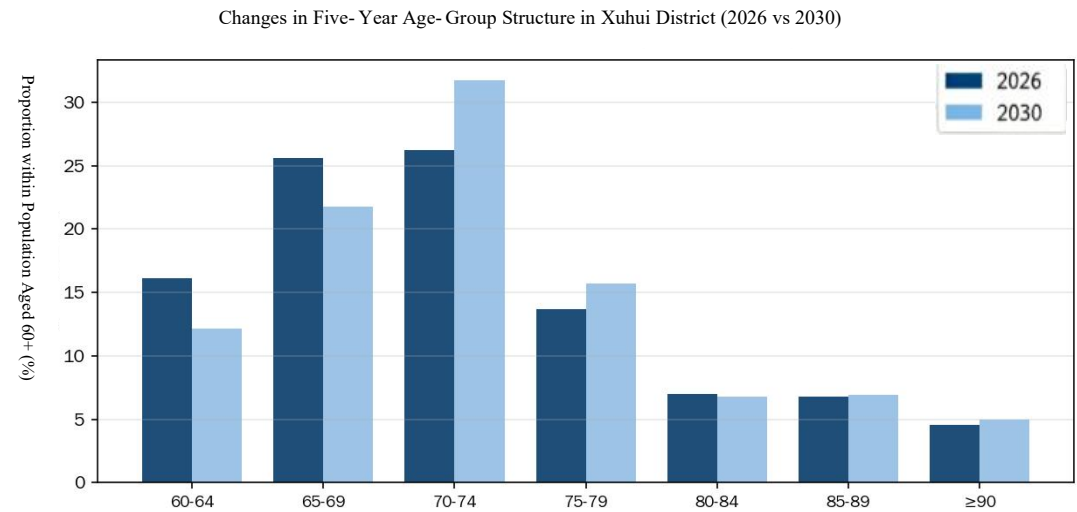


Figure 2 Structural Changes in Five-Year Age Groups (2026 vs 2030)

Two structural highlights: First, **the 70–74 age group grows fastest**, reflecting the baby boomer generation advancing into older ages; second, the sum of the three groups aged 80 and above (80–84, 85–89, ≥ 90) rises steadily, which is precisely the source supporting the rise in the advanced aging ratio. The 60–64 group declines moderately, continuing the real trajectory of the observation period without the abnormal "collapse" seen in simple methods.

VIII. Uncertainty Expression (I): $\pm 1\sigma$ Error Band

8.1 What It Expresses

The forecast of any model is not an absolutely precise point, but carries a certain margin of error. The question answered by the $\pm 1\sigma$ error band is: **"In terms of the model's own fitting accuracy alone, how much will this predicted value roughly fluctuate up and down?"** It is synthesized from two error sources:

- **Calibration error:** The Step 1 conversion relationship itself does not perfectly fit historical data, leaving residuals.
- **Extrapolation error:** Step 3 extrapolating Shanghai data into the future also introduces deviations.

The two errors are synthesized into $\pm 1\sigma$ for each forecast value through standard "error propagation" rules. σ (sigma) is the statistical symbol for "standard deviation", and $\pm 1\sigma$ roughly corresponds to "about two-thirds certainty of falling within this interval".

8.2 $\pm 1\sigma$ Band for Each Threshold in 2030

Table 5 $\pm 1\sigma$ Error Band for Nine Thresholds in Xuhui District in 2030 (10,000 Persons)

Threshold	Baseline (10,000)	-1σ Lower Bound	$+1\sigma$ Upper Bound	$\pm\sigma$ (10,000)
60+	36.694	36.499	36.889	0.1948
65+	32.243	32.089	32.397	0.1544
70+	24.260	24.103	24.417	0.1572

75+	12.606	12.462	12.750	0.1438
80+	6.849	6.739	6.960	0.1105
85+	4.348	4.209	4.487	0.1391
90+	1.817	1.678	1.956	0.1386
95+	0.421	0.282	0.559	0.1385
100+	0.046	-0.093	0.184	0.1385

8.3 What $\pm 1\sigma$ Includes and Excludes

Includes: Step 1 calibration residuals + Step 3 extrapolation residuals, synthesized via error propagation.

Excludes: Possible directional errors in extrapolation assumptions themselves, structural fluctuations in five-year groups, functional form risks of age-ratio functions, and future unforeseen public emergencies (such as new epidemics). These belong to risks "outside the model framework" and need to be supplemented by the scenario intervals in the next section and limitations in Section 10.

IX. Uncertainty Expression (II): Three-Scenario Interval

9.1 What It Expresses

$\pm 1\sigma$ measures "how accurate the model is", while the three-scenario interval measures something else: **"Future population growth may be faster or slower, and how wide is the high-low interval brought about by this?"** It is not a model error, but a characterization of "uncertainty in future growth paths", suitable for external reporting and graphical presentation.

9.2 How High and Low Scenarios Are Determined

Based on **the historical annual growth rates** of the 60+ population in Shanghai from 2017 to 2022, we calculate its average growth rate and fluctuation range (standard deviation), and construct one lower and one higher growth scenario accordingly:

Table 6 Growth Rate Parameters for Three Scenarios (Based on Historical Growth Rate of Shanghai 60+)

Parameter	Value	Meaning
Average Growth Rate μ	2.7462%	Historical average annual growth rate of 60+
Fluctuation Range σ	0.9324%	Standard deviation of growth rate
Low Scenario Growth Rate	0.9187%	Mean deviates downward by about two standard deviations
High Scenario Growth Rate	4.5736%	Mean deviates upward by about two standard deviations

9.3 Scaling with "Unified Multiplier" to Ensure Scenario Order Integrity

To avoid compounding growth repeatedly, we convert scenario growth rates into a **"dynamic multiplier"** relative to baseline. The further the year, the more the multiplier deviates from 1. The key design is: within the same year and same scenario, **all age thresholds and five-year groups use the exact same multiplier**. Doing so guarantees that the order "Low Scenario < Baseline < High Scenario" holds strictly across every age bracket without inversions. Taking 2030 as an example, the low scenario multiplier is **0.9142**, and the high scenario multiplier is **1.0921**.

9.4 Relationship Between the Two Uncertainties

The two measure different objects and cannot be merged; they should be presented side-by-side and explained separately. $\pm 1\sigma$ expresses "fitting error of the baseline model" for rigorous technical verification; three scenarios express "up-and-down deviations of future growth paths" for planning stress testing and external communication. It is recommended to simultaneously provide baseline, three-scenario intervals, and $\pm 1\sigma$ bands in reports, noting their meanings separately.

X. Accompanying Data File Guide (17 Worksheets)

The accompanying Excel file is organized under the logic of "Methodology Description → Step-by-Step Calculation → Uncertainty", facilitating level-by-level tracing of the origin of each figure:

Table 7 List and Functions of Worksheets

Worksheet	Content and Function
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1 Instructions and Methodology	Complete text description of data sources, three-step method, uncertainties, and limitations
2 Step 1 Regression Model	Formulas, coefficients, and goodness of fit of five conversion relationships
3 Step 1 Fit Diagnostics	Year-by-year comparison and residuals between conversion predicted values and real observed values
4 R(a) Age-Ratio Function	Ratio of "Xuhui ÷ Shanghai" proportions across years and age points
5 Shanghai 2017-2030	Population and proportions across nine thresholds in Shanghai (observed + extrapolated, color-coded)
6 Xuhui District 2019-2025	Nine thresholds in Xuhui historical period (three observed + six completed by R(a))
7 Xuhui District 2026-2030 Cumulative	Cumulative population distribution across nine thresholds during forecast period (Baseline)
8 Xuhui District 2026-2030 Age Groups	Population and proportions of seven five-year age groups during forecast period
9 Reference Birth and Death Rates	Birth rates and death rates in Xuhui (explaining causes of pandemic trough)
10–13 $\pm 1\sigma$ Error Band	Methodological description and error bands for cumulative thresholds / five-year groups / key indicators
14–17 Three-Scenario Interval	Methodological description and three scenarios for cumulative thresholds / five-year groups / summary table